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Quantifying Uncertainty in AI Visibility

A Statistical Framework for Generative Search Measurement

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Abstract

AI-powered answer engines are inherently non-deterministic: identical queries submitted at different times can produce different responses and cite different sources. Despite this, current approaches to measuring domain visibility in generative search rely on single-run point estimates of citation share and prevalence, treating them as fixed values. This paper argues that citation visibility metrics should be treated as sample estimators of an underlying response distribution. We conduct an empirical study of citation variability across three generative search platforms—Perplexity Search, OpenAI SearchGPT, and Google Gemini—using repeated sampling across three consumer product topics under two sampling regimes: daily collections over nine days and high-frequency sampling at ten-minute intervals. Citation distributions follow a power-law form and exhibit substantial variability across repeated samples. Bootstrap confidence intervals reveal that many apparent differences between domains fall within the noise floor of the measurement process. Distribution-wide rank stability analysis further demonstrates that citation rankings are unstable across samples, not only among top-ranked domains but throughout the frequently cited domain set. These findings show that single-run visibility metrics provide a misleadingly precise picture of domain performance in generative search. We argue that citation visibility must be reported with uncertainty estimates and provide practical guidance on sample sizes required to achieve interpretable confidence intervals.

CCS Concepts: Information systems → Information retrieval → Evaluation of retrieval results; Information systems → World Wide Web → Web searching and information discovery.

Additional Keywords and Phrases: AI visibility, answer engine optimization, generative engine optimization.

1 Introduction

Generative search engines have transformed how users encounter information online. Rather than returning a ranked list of hyperlinks, platforms such as Perplexity Search, OpenAI SearchGPT, and Google Gemini synthesize a natural-language response to each query and attribute claims within that response to a set of cited web sources. For brand managers, digital marketers, and content strategists, these citations have become a new form of visibility: appearing in a cited source confers authority and drives traffic in a way that a traditional search ranking no longer guarantees.

Practitioners have responded by developing measurement frameworks that report, for example, citation counts, citation share, and citation prevalence as metrics of AI visibility. These reports typically derive from a single batch of queries submitted on a single occasion. The resulting numbers are treated as ground truth: domain A has 12% citation share, domain B has 8%, and A is therefore

more visible. Optimization campaigns are launched, interventions are measured, and conclusions are drawn—all on the basis of single-run point estimates.

This paper argues that such measurements are fundamentally unreliable, because the systems they measure are non-deterministic. The same query submitted twice to the same platform can produce a different cited source set. This is not a bug or an edge case: it is a design property of generative AI systems, which use stochastic sampling at generation time. The practical consequence is that citation visibility metrics are random variables, not fixed values, and any single measurement of them carries unquantified uncertainty that may be large enough to invalidate common analytical conclusions.

1.1 Motivating Example

Consider a concrete decision a practitioner faces every day. A brand manager measuring domain visibility on SearchGPT for the topic of running gear collects a sample of 200 queries and observes that *tomsguide.com* has a citation share of approximately 9.5%, while *runnersworld.com* has a citation share of approximately 6.0%. The conclusion seems clear: Tom's Guide is the dominant source for this topic on this platform, outperforming Runner's World by more than three percentage points. Resources are allocated accordingly.

The 95% bootstrap confidence intervals tell a different story. The interval for *tomsguide.com* spans approximately 5.5% to 12.5%; the interval for *runnersworld.com* spans approximately 4.0% to 8.0%. These intervals overlap substantially: the entire *runnersworld.com* interval falls within the *tomsguide.com* interval. The apparent difference of 3.5 percentage points is well within the range of sampling noise. A practitioner cannot conclude from this sample that *tomsguide.com* outperforms *runnersworld.com* with any statistical confidence.

This is not a pathological edge case. Across the platforms and topics studied in this paper, overlapping confidence intervals of this kind are the norm rather than the exception for domains that appear to differ in citation share by less than 5-7 percentage points. The bootstrap confidence interval makes the uncertainty explicit and actionable: it transforms an apparently decisive ranking into a result that is statistically indistinguishable from a tie. Citation visibility metrics must be accompanied by uncertainty estimates, or they are liable to mislead.

This paper reframes the measurement of domain visibility in generative search as a statistical estimation problem for stochastic information systems and demonstrates empirically that reliable visibility measurement requires repeated sampling and explicit uncertainty quantification.

1.2 Contributions

This paper makes the following contributions:

- An empirical characterization of citation variability in three generative search platforms across repeated sampling, demonstrating that stochastic system behavior produces material differences in measured domain visibility.
- A formal framework distinguishing system-level stochasticity (the answer engine produces different outputs on repeated calls) from measurement uncertainty (estimates derived from finite samples are imprecise) and showing that both are present and consequential.
- A demonstration that citation count distributions follow a power-law form, and that variability differs structurally between the head and tail of the distribution—with implications for how frequently and infrequently cited domains should be analyzed.

- An application of bootstrap confidence intervals to citation visibility metrics, showing that they are both necessary (intervals are wide relative to observed differences) and computationally tractable.
- A distribution-wide rank stability analysis using weighted Spearman rank correlation, demonstrating that citation rank instability extends across the full frequently-cited domain set and is not confined to the top-ranked domains, establishing that confidence intervals are necessary for inference at any rank position in the distribution.
- A methodological control using content checksums to verify that observed variability is attributable to the answer engines rather than to changes in the content of cited pages.

2 Related Work

2.1 Citation Quality in Generative Search

A growing body of work examines the accuracy and faithfulness of citations in generative search outputs. Liu et al. establish that generative systems frequently cite sources that do not support the attributed claims and introduce metrics for citation recall and precision [12]. Gao et al. present the ALCE benchmark for automatic citation evaluation, which operationalizes citation quality as the degree to which cited passages entail generated claims [7]. Xu et al. extend this framework with CiteBench, introducing fine-grained criteria for evaluating citation faithfulness across multiple dimensions [15]. Li et al. propose AttributionBench, targeting the attribution verification problem specifically [11].

This literature treats citation accuracy as the central measurement challenge: given a response and its citations, do the citations support the claims? Implicit in this framing is that each measurement is a reliable observation about a well-defined system state. None of these works address the variability of citation selection across repeated runs of the same query—the question of whether the set of cited sources itself is stable. Our work complements this literature by characterizing the stochastic properties of the citation process rather than the quality of individual citations.

2.2 Visibility and Optimization in Generative Search

Aggarwal et al. introduce the GEO framework and the GEO-BENCH evaluation suite, which defines domain visibility metrics and reports the effects of content interventions on citation rates across answer engines [1]. GEO is the most directly competitive work to ours: it defines visibility metrics that align closely with citation share and citation prevalence as we define them, and it reports statistically significant improvements from content interventions.

The critical gap is the absence of confidence intervals on both the baseline and post-intervention measurements. Without uncertainty quantification, it is impossible to determine whether reported visibility improvements exceed the noise floor of the measurement process itself. Our work demonstrates that the noise floor is non-trivial: CI widths of 5-7 percentage points on citation share are common for SearchGPT domains, and improvements of this magnitude or smaller cannot be reliably attributed to an intervention without repeated sampling and statistical validation.

2.3 User Behavior and Attention in AI-Cited Responses

The behavioral consequences of citations in AI-generated responses represent an adjacent research thread. Joachims et al. [10] and Dupret et al. [5] establish the foundational attention and click bias

models for traditional search, which inform how citation position may influence user engagement. Cutrell et al. document the role of snippet quality in click behavior [3]. More recently, Ding et al. examine trust effects of citations in AI responses [4], and He et al. study citation UI design and its effects on user attention [8]. Venkit et al. characterize user behavior patterns with answer engines more broadly [14].

Bink et al. investigate credibility perceptions of featured snippets [2], and Hu et al. examine inconsistency in AI Overview presence across repeated queries [9]. The latter is the empirically closest work to ours: they observe a 33% inconsistency rate in whether AI Overviews appear at all for a given query, which is an instance of the same system-level stochasticity we study. However, Hu et al. treat inconsistency as a confound to be controlled rather than as the object of study; we treat it as the primary phenomenon of interest.

2.4 Evaluation Variance and Statistical Rigor

Madaan et al. address variance in Large Language Model (LLM) benchmark evaluation, arguing that single-number comparisons without variance reporting are misleading, and introducing variance metrics for benchmark results [13]. This work is the closest methodological ally to ours: both papers argue that point estimates without uncertainty quantification are insufficient, and both use sampling-based methods to characterize variability.

The key distinction is scope: Madaan et al. address evaluation variance in a controlled benchmark setting, where queries and evaluation criteria are fixed and the only source of variability is the model's stochastic generation. We address measurement uncertainty in a deployed, production search system where ground truth is unavailable, the response distribution is non-stationary, and sampling is the only feasible approach to characterizing system behavior. The bootstrap methods we apply follow the tradition of Efron and Tibshirani [6] and are standard in empirical software engineering and information retrieval evaluation.

3 Background and Definitions

3.1 Generative Search and Citation Structure

An answer engine is a system that accepts a natural-language query and returns a synthesized natural-language response, where claims within the response are attributed to a set of cited web sources. We use the terms answer engine and generative search engine interchangeably throughout. Each platform studied here—Perplexity Search, OpenAI SearchGPT, and Google Gemini—implements this paradigm, though with meaningful differences in citation volume, citation presentation, and response format.

A response is a single synthesized answer to a single query. Each response contains one or more citations, where a citation is a reference to a specific URL that the platform identifies as a source supporting the response content. The domain of a citation is the registered domain of the cited URL (e.g., *nationalgeographic.com* for any URL under that domain). Citation extraction is platform-specific: Perplexity and SearchGPT expose citations as structured fields in their API responses (Appendix A provides illustrative examples), while Gemini embeds source references in a format requiring text-level parsing. All three platforms were handled by platform-specific extraction procedures; see Section 4.4 for details.

3.2 Visibility Metrics

We define three visibility metrics. Let S be a sample of N responses to a set of queries submitted to a single platform. Let $c(d,\ S)$ denote the total number of times domain d is cited across all responses in S, and let

$$C(S) = \sum_{d} c(d, S) \tag{1}$$

be the total number of citations in S.

Citation count for domain d in sample S is $c(d,\ S)$. Citation count is an unbounded sum that scales with N and with the per-response citation volume of the platform. It is not directly comparable across platforms or across samples of different sizes.

Citation share for domain d in sample S is

$$s(d, S) = \frac{c(d, S)}{C(S)}. \tag{2}$$

Citation share normalizes by total citations, making it comparable across samples and across platforms. The near-perfect correlation between citation count and citation share within a platform (Spearman $\rho = 0.994$-0.997; Section 5.1) confirms that share preserves the ordinal information of count while enabling cross-platform comparison.

Citation prevalence for domain d in sample S is

$$p(d, S) = \frac{\left|\{\, r \in S : d \text{ cited in } r \,\}\right|}{N}. \tag{3}$$

Prevalence measures the fraction of responses that include at least one citation to d, regardless of how many times d is cited within a single response, r. Prevalence is sensitive to the breadth of domain coverage across responses rather than the depth of citation within any single response.

In industry practice, metrics such as citation count, citation share, and citation prevalence are typically reported as fixed values describing domain visibility on a platform. This interpretation is misleading. Each metric is computed from a finite sample of responses and therefore represents an estimate of an underlying response distribution rather than a deterministic property of the system. Formally, these quantities should be interpreted as sample estimators of population parameters governing the probability that a domain is cited by the platform. As a result, the observed values depend on the particular set of responses collected and are subject to sampling variability.

All three are therefore sample statistics, functions of a finite collection of responses, and therefore random variables subject to sampling variation.

Formally, citation share can be written as the estimator

$$\widehat{s}_S(d) = \frac{c(d, S)}{C(S)} \tag{4}$$

where $\widehat{s}_S\,(d)$ denotes the estimator of citation share for domain d computed from response sample S. Under this interpretation, $\widehat{s}_S\,(d)$ estimates the expected citation share of domain d under repeated sampling of responses from the platform.

This interpretation places citation visibility measurement within the broader framework of statistical estimation for stochastic systems. A generative search engine can be viewed as producing responses drawn from an underlying response distribution conditioned on a query. Visibility metrics computed from a finite set of queries therefore estimate properties of that distribution rather than fixed attributes of the platform. Reliable measurement requires repeated sampling and uncertainty

quantification to characterize the variability of these estimates. The analyses in this paper adopt this perspective, treating citation visibility metrics as estimators whose statistical properties can be studied through repeated sampling.

This perspective reframes AI visibility measurement as a statistical estimation problem for stochastic information systems rather than a deterministic ranking analysis.

3.3 System-Level Stochasticity vs. Measurement Uncertainty

Because visibility metrics are sample estimators computed from a finite collection of responses, their observed values can vary across repeated measurements. Two distinct sources of variability contribute to this variation. The first arises from the non-deterministic behavior of generative search systems themselves, while the second arises from the statistical uncertainty inherent in estimating population quantities from finite samples. We distinguish these sources as system-level stochasticity and measurement uncertainty, respectively.

Two distinct sources of variability are present in this study, and they must not be conflated.

System-level stochasticity refers to the property that the answer engine itself is non-deterministic: the same query submitted twice produces different responses, because generation is stochastic (temperature > 0), and because the retrieval components that ground citations may return different candidate documents on each call. This is a property of the system, not of the measurement process.

Measurement uncertainty refers to the property that any finite sample of responses produces estimates with sampling error. Even if the underlying response distribution were perfectly stable, a sample of 200 queries would produce citation share estimates that deviate from the true population values. This is a property of the measurement process.

Both are present in this study. The high-frequency sampling regime, queries submitted at ten-minute intervals, helps isolate system-level stochasticity by minimizing the opportunity for content changes in cited pages, for example, between samples. Bootstrap confidence intervals address measurement uncertainty by characterizing the sampling distribution of the estimates. The content-change validation in Section 6 provides empirical evidence that the variability observed is predominantly system-level: the underlying pages are largely stable, and the citation variability reflects engine behavior rather than changing source material.

A third property, within-sample non-stationarity, is distinct from both. A citation distribution is stationary if its statistical properties (mean, variance, and correlation structure) remain constant across the sequence of queries within a single collection run. Non-stationarity means the distribution shifts across the query sequence: early queries in the batch elicit a systematically different citation pattern than later queries, so the 200 responses in a single sample are not exchangeable draws from a fixed distribution. The practical consequence is that confidence intervals built on the stationarity assumption, including the bootstrap, may understate true uncertainty, and convergence of CI width with sample size may be non-monotone. Within-sample non-stationarity is most likely driven by query-ordering effects: if queries of similar type cluster together in the submission sequence, the running citation distribution shifts as the composition of the query batch changes. This is a measurement design problem rather than a system property, and it is distinct from between-sample stationarity, whether the citation distribution is the same on different occasions, which is what the daily time series in Section 5.2 and the rank stability analysis in Section 5.8 address. Section 5.7 provides empirical evidence of within-sample non-stationarity for SearchGPT, in the form of non-monotone CI convergence curves.

The experimental design in Section 4 is structured to isolate these two sources of variability by combining repeated sampling with short-interval collections that minimize the probability of content changes in cited sources.

4 Experiment Design

4.1 Topics

Three consumer product topics were selected to represent varied market structures and information environments. Bird feeders is a fragmented market with no dominant brand, driven by ecological diversity and hobbyist communities; the information environment includes niche specialty sites alongside general natural history publishers. Running gear is a brand-concentrated, multi-product category spanning footwear, electronics, and apparel; the information environment mixes commercial retailers, specialist review sites, and general sports media. Multivitamins for adults is a health and life sciences context with strong representation from institutional and clinical sources alongside commercial and wellness publishers. All three topics are unbranded queries of broad consumer interest.

4.2 Query Generation

For each topic, queries were generated by prompting an LLM (ChatGPT) to produce queries conditioned on a randomly selected query type drawn from a fixed set of ten categories: strengths and weaknesses, popular options, value for money, environmental impact, expert recommendations, feature comparisons, customer satisfaction, what to look for, common complaints, and emerging trends. This process was repeated until 200 queries had been generated for each topic. The ten-category structure is confirmed empirically: facet mapping across all collected responses yields exactly ten distinct query type values.

Uniqueness was not enforced. The intent was to simulate the distribution of queries that an answer engine receives from a diverse user population, in which popular queries recur. If the generating model produced a repeated query, it was retained on the grounds that frequently generated queries are likely to correspond to frequently asked user queries. Table 1 summarizes the resulting query sets, including total queries, unique queries, and repetition rate per topic.

Table 1: Query generation summary: total queries, unique queries, and repetition rate per topic.

Topic	Total Queries	Unique Queries	Repetition Rate
bird feeders	200	172	14.0%
multivitamins for adults	200	152	24.0%
running gear	200	185	7.5%

We acknowledge that query generation by LLM introduces a potential confound: the generating model may have implicit associations between topics and sources that shape the query distribution. This is discussed further in Section 8.

4.3 Data Collection

The same 200 queries per topic were submitted to Perplexity Search, OpenAI SearchGPT, and Google Gemini under two sampling regimes.

Daily samples: queries were submitted once per day over nine consecutive days, yielding nine samples of approximately 200 responses per topic per platform. Table 2 provides summary statistics for the responses and citations by platform and topic. (A daily breakdown of these same statistics is available in Appendix B.)

Table 2: Summary statistics for the daily sampling regime: responses and citations by topic and platform.

			Citations				
Platform	Topic	Mean Responses	Mean	Median	Std	Min	Max
Gemini	bird feeders	198.8	42.6	37.0	22.7	6	158
Gemini	multivitamins	198.6	39.6	36.0	19.6	3	197
Gemini	running gear	199.0	43.1	40.0	21.0	1	202
SearchGPT	bird feeders	196.2	7.1	7.0	2.7	1	18
SearchGPT	multivitamins	193.9	5.9	5.0	2.0	1	15
SearchGPT	running gear	188.1	6.6	5.0	2.6	1	19
Perplexity	bird feeders	199.8	22.0	20.0	9.7	2	72
Perplexity	multivitamins	199.9	20.0	19.0	8.3	3	52
Perplexity	running gear	199.8	22.5	22.0	8.7	4	56

The substantial variation in median citation volume, from 5-7 citations per response on SearchGPT to 36-40 on Gemini, is immediately visible and has direct implications for how metrics are defined and compared across platforms.

High-frequency samples: for the running gear topic, queries were submitted to Gemini, SearchGPT, and Perplexity at ten-minute intervals over approximately four hours, yielding 25 samples per platform. The goal was to minimize the probability of content changes in cited pages between samples, thereby isolating system-level stochasticity. At ten-minute intervals, substantive editorial changes to web content are unlikely for the majority of sources. Table 3 provides summary statistics for the responses and citations by platform.

Table 3: Summary statistics for the high-frequency sampling regime: responses and citations by platform.

			Citations				
Platform	Topic	Mean Responses	Mean	Median	Std	Min	Max
Gemini	running gear	198.8	47.1	42.0	23.9	1	213
SearchGPT	running gear	190.8	6.7	5.0	2.7	1	40
Perplexity	running gear	199.9	22.7	22.0	8.8	3	63

4.4 Citation Extraction

Citations were extracted from raw generative responses using platform-specific parsing procedures, reflecting differences in how each platform structures its outputs. Extracted fields include the reference URL and domain for each cited source, along with the response and query identifiers necessary for linking citations back to their generating responses.

After extraction and deduplication, the daily dataset contained 374,052 citations and the high-frequency dataset contained 379,276 citations. The extraction methodology was consistent across all samples for a given platform.

4.5 Content-Change Monitoring

As a methodological control, the HTML of cited URLs was scraped and stored during each data collection job. Human-readable content was extracted from raw HTML using Trafilatura, and a SHA-256 checksum was computed over the extracted text. Checksums were stored alongside scraping metadata in a database indexed by URL and job identifier, enabling detection of content changes between samples without requiring re-retrieval and re-comparison of full HTML. Scraping coverage is subject to server-side blocking (e.g., Cloudflare, CAPTCHA challenges) and to format limitations (video content and PDF documents, for example, cannot be processed by Trafilatura). Results of the content-change validation are reported in Section 6.

5 Findings

The findings are organized from individual responses to aggregate visibility metrics. We first show that variability is visible at the response level (Section 5.1), then characterize aggregate visibility over time (Section 5.2). Next, we differentiate between frequently and infrequently cited domains and establish how they will be handled in this analysis (Section 5.3). We establish the distributional foundation, power-law structure (Section 5.4) and log-space dispersion (Section 5.5), before demonstrating the application of bootstrap confidence intervals (Section 5.6) and analyzing the effect of sample size on interval width (Section 5.7). Finally, we close with a distribution-wide assessment of rank stability across samples (Section 5.8).

5.1 Variability in Individual Responses

Before aggregating across queries, we examine whether variability is visible at the individual response level. For the same query submitted across multiple samples, the answer engines returned different sets of cited sources. Table 4 summarizes the response-level citation overlap structure: rows represent the mode number of citations per response for each platform, and columns represent the count of matching citations across repeated runs of the same query. Matching is measured at the domain level (not URL level): two responses match on a domain if both cite at least one URL from that domain. URL-level matching would be strictly more variable

Table 4: Response-level citation matching. Rows give per-platform Jaccard overlap statistics across repeated runs of the same query, including identical citation rate, zero-overlap rate, and median Jaccard index. Matching is at the domain level.

Platform	Topic	Queries	Pairs	Median Jaccard	Identical Rate	Zero Overlap Rate	Mean Intersection	Mean Unique Domains
Gemini	bird feeders	189	6,804	0.31	0.07	0.35	4.90	10.54
Gemini	multivitamins	192	6,912	0.29	0.01	0.80	4.49	10.55
Gemini	running gear	192	6,912	0.29	0.10	1.62	4.73	10.88
SearchGPT	bird feeders	180	6,480	0.40	7.25	6.08	2.50	4.38
SearchGPT	multivitamins	180	6,480	0.38	8.02	7.19	2.03	3.76
SearchGPT	running gear	169	6,084	0.33	5.34	8.93	2.04	4.04
Perplexity	bird feeders	198	7,128	0.50	3.93	2.40	3.32	5.24
Perplexity	multivitamins	199	7,164	0.50	4.65	1.05	3.34	5.11
Perplexity	running gear	198	7,128	0.50	3.25	1.07	3.49	5.31

Platform-level Jaccard medians are 0.29-0.31 for Gemini, 0.33-0.40 for SearchGPT, and 0.50 for Perplexity across all three topics. The identical citation rate, where two responses to the same query share every cited source, ranges from near-zero for Gemini (0.01-0.10%) to 3-8% for SearchGPT and Perplexity. The zero-overlap rate is highest for SearchGPT (6-9%) and lowest for Perplexity (1-2%), consistent with the bimodal behavior shown below.

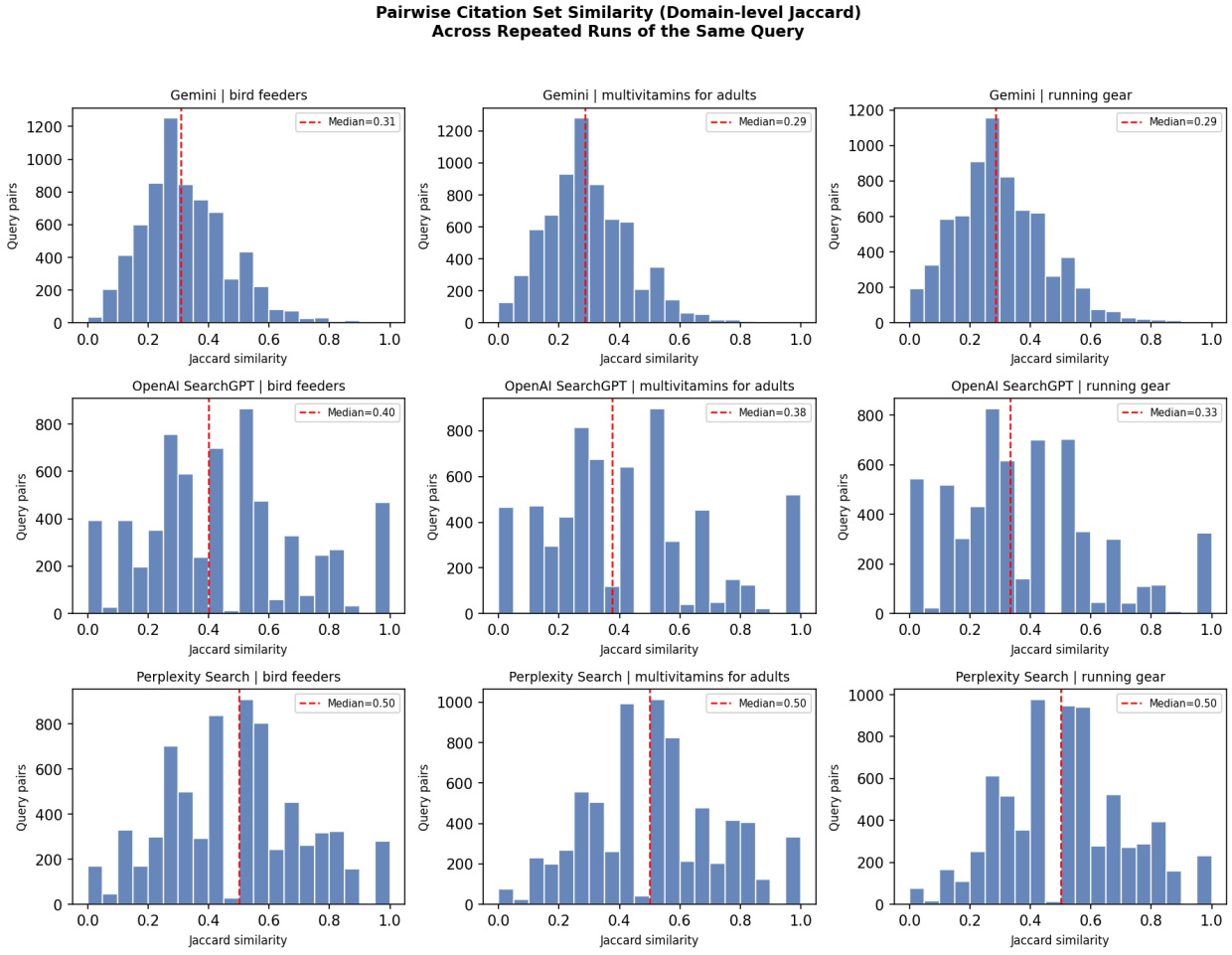


Figure 1: Pairwise citation set similarity (domain-level Jaccard) across repeated runs of the same query: rows = platform, columns = topic.

The pairwise Jaccard distribution histograms reveal the structure behind these summaries. Gemini's distributions are broad and unimodal, peaking near 0.30-0.35 with a slight skew to the right. The platform rarely produces either complete overlap (Jaccard = 1) or complete divergence (Jaccard = 0).

The SearchGPT distributions are substantially broader and more irregular than Gemini's. Rather than forming a single unimodal peak, the histograms span nearly the full 0-1 range and exhibit multiple local peaks. This pattern indicates that repeated runs of the same query frequently produce distinct citation sets, with similarity varying from minimal overlap to near identity. The multimodal structure suggests regime-like behavior in the citation distribution rather than sampling from a single stable pool of sources.

The Perplexity distributions occupy an intermediate position between Gemini and SearchGPT. They are centered at substantially higher similarity levels (median ≈ 0.50 across topics), indicating that repeated runs tend to share roughly half of their cited domains. The distributions remain moderately broad and show mild multimodality, suggesting that Perplexity responses draw from a stable core of frequently cited domains while alternating among a small number of secondary citation sets. This behavior contrasts with Gemini's smoother unimodal distributions centered near 0.30 and with SearchGPT's highly irregular, multi-peaked distributions that span nearly the full 0-1 range. A second analysis plots median Jaccard against the modal citation count per response within each platform.

The number of citations in a response might plausibly affect citation-set similarity. In principle, responses that contain more citations provide more opportunities for divergence across runs: each additional citation could be replaced by a different source, potentially reducing overlap between citation sets. If this mechanism dominated, Jaccard similarity would decline as the number of citations per response increases.

Figure 2 shows that this effect is largely absent. Across all three platforms and topics, Jaccard similarity remains essentially flat across the observed range of citation counts. Gemini maintains similarity near 0.30 across responses containing roughly 10 to 70 citations. SearchGPT remains near 0.40-0.42 across 3 to 12 citations, while Perplexity holds near 0.50 across 5 to 35 citations. The fitted lines show little systematic slope, and the uncertainty bands overlap broadly across citation-count bins.

The stability of these curves indicates that citation-set similarity is largely independent of the number of citations in a response. Responses with many citations are not materially more divergent across runs than responses with few citations. Instead, the dominant determinant of citation consistency is platform identity. Each system exhibits a characteristic level of overlap across repeated runs (approximately 0.30 for Gemini, 0.40 for SearchGPT, and 0.50 for Perplexity), regardless of how many sources appear in individual responses.

This result suggests that citation variability arises primarily from platform-level retrieval and ranking behavior, rather than from simple combinatorial effects associated with response length. In other words, larger citation sets do not inherently produce more unstable citation patterns; the consistency of those sets is determined mainly by how each system constructs and selects its evidence base.

Importantly, this pattern holds despite substantial variation in citation counts within each platform. If citation-set similarity were primarily driven by the combinatorics of set size, we would expect similarity to increase with the number of citations per response. The absence of such a trend indicates that the observed differences in similarity are not a mechanical artifact of response length but instead reflect platform-specific citation selection behavior.

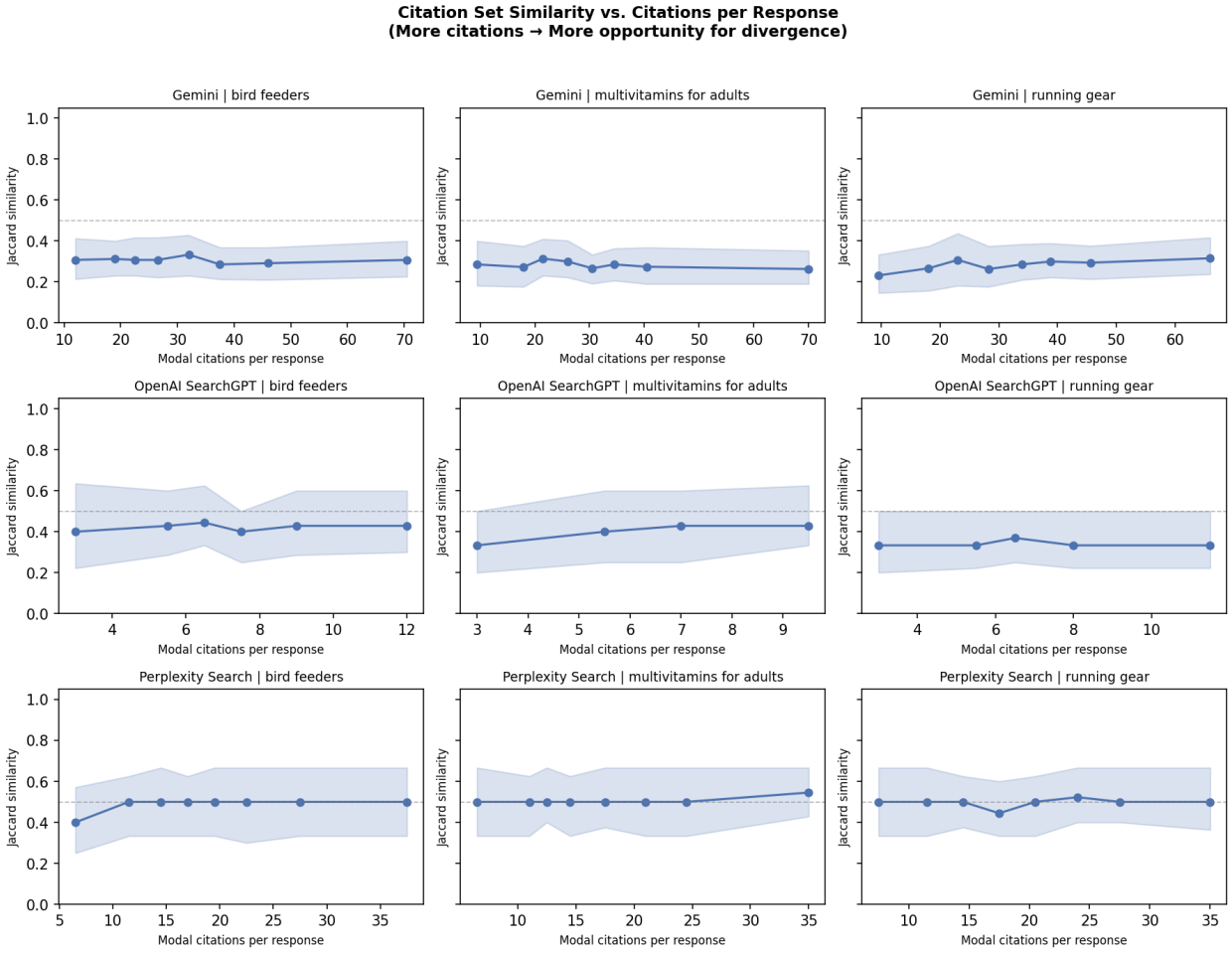


Figure 2: Citation set similarity vs. citations per response: rows = platform, columns = topic. Jaccard similarity is largely independent of the number of citations.

5.2 Aggregate Visibility Metrics

We now move from individual responses to aggregated visibility metrics across all queries in a sample. Figure 3 shows the high-frequency time series for the running gear topic, plotting citation share (top row) and citation prevalence (bottom row) for the top-ranked domains on each platform across consecutive ten-minute samples.

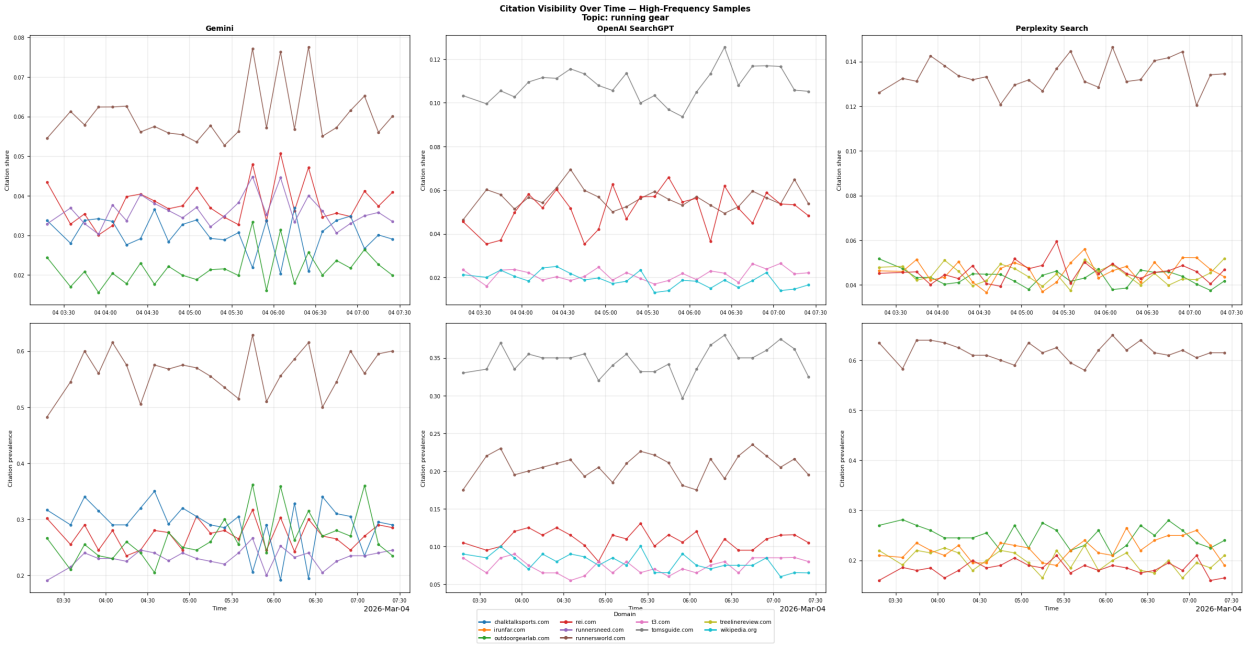


Figure 3: Citation visibility over time: high-frequency samples. Topic: running gear. Top row: citation share; bottom row: citation prevalence. Left: Gemini (25 samples, 2026-Mar-04); center: SearchGPT (25 samples, same window); right: Perplexity (25 samples, same window). Top-ranked domains shown as colored lines.

Because this window is far too short for meaningful changes in the underlying web content, variation in these series reflects instability in the platforms' citation selection rather than changes in the source material itself.

The three platforms exhibit distinct stability regimes. On Gemini, *runnersworld.com* is the most cited domain, with citation share fluctuating between 5.3% and 7.7%. The remaining top domains exhibit substantial short-interval volatility, particularly in the latter half of the series where citation shares oscillate sharply. The prevalence series mirrors this pattern: several domains show abrupt rises and drops over adjacent samples, indicating frequent reshuffling of the citation set rather than smooth sampling from a stable distribution.

On SearchGPT, *tomsguide.com* is the top-cited domain with a citation share of roughly 9.4-12.5%, which remains moderately stable across the sample window. Instability appears primarily among the lower-ranked domains, which tend to swap positions in pairs—most visibly between the second and third and fourth and fifth positions. This pattern suggests that SearchGPT maintains a relatively stable leading source while the remainder of the citation set alternates among a small group of competing domains.

Perplexity shows the most stable behavior overall. *runnersworld.com* again appears as the dominant domain with citation share between 12.0% and 14.6%, and both the share and prevalence series remain comparatively smooth across the window. The lower-ranked domains cluster at similar

citation levels and occasionally exchange rank order, but the amplitude of these fluctuations is small relative to the other platforms. This pattern suggests a stable core citation set with minor rotation among secondary sources.

Taken together, the platforms form a clear ordering in short-interval stability: Perplexity is the most stable, followed by SearchGPT, with Gemini exhibiting the greatest volatility. This ordering mirrors the patterns observed in the daily sampling regime and in the log-space dispersion analysis presented in subsequent sections. The consistency of this ranking across time scales indicates that the stability differences are structural properties of each engine's retrieval and citation-selection process, rather than artifacts of the sampling procedure.

The prevalence series closely track the behavior seen in citation share, confirming that the volatility reflects changes in which domains appear in responses rather than small fluctuations in citation counts within otherwise stable citation sets.

Figure 4 presents the daily time series across all nine topic-platform panels, confirming that the patterns seen in the high-frequency window persist at daily resolution and over the nine-day window.

A key observation is that domain rankings by citation share are unstable across samples. Here, the most anomalous behavior is *nationalgeographic.com* in the Gemini bird feeders panel: the domain registers a citation share of approximately 0.032 on day 1 and reverts to near-zero for all subsequent days. In the SearchGPT multivitamins panel, yahoo.com citation share ranges from approximately 0.092 to 0.160 within the nine-day window—a factor of nearly 2× for the top-ranked domain. The Perplexity bird feeders panel shows *allaboutbirds.org* swinging between 0.099 and 0.134 across the nine days but starting and ending at 0.124.

Figure 5 quantifies the relationships among the three visibility metrics across all samples.

Citation count and citation share are nearly perfectly correlated within platform ($\rho = 0.958$-0.997), confirming that share preserves the ordinal information of count while enabling cross-platform comparison. The x-axis scales differ substantially across platforms (Gemini/Perplexity: 0-700 citations; SearchGPT: 0-175), making the volume difference immediately visible and confirming that raw counts cannot be compared cross-platform. Citation prevalence is moderately correlated with both count and share ($\rho = 0.756$-0.833), with substantial scatter at higher values. The fan-shaped scatter in the share-prevalence panels confirms that prevalence is structurally distinct from share and captures a different dimension of visibility: a domain can have high share by appearing many times in a few responses, or high prevalence by appearing once in many responses.

As shown in Table 2, median citations per response differ substantially across platforms: 36-40 for Gemini, 19-22 for Perplexity, and 5-7 for SearchGPT. These volume differences mean that raw citation counts are not comparable across platforms; citation share is the appropriate primary metric for cross-platform analysis.

5.3 Infrequently Cited Domains

Domains that are not cited in every sample present a distinct measurement challenge. When a domain is absent from a sample, its measured visibility is zero, but this does not imply that its citation probability is zero. The domain may simply not have been sampled in that run. Standard bootstrap methods designed for continuously distributed data do not apply directly to these zero-inflated observations.

For the purposes of this analysis only, a domain is classified as frequently cited if it appears in every sample for a given platform and topic. Figure 6 plots the empirical distribution of domain appearance counts across the nine daily samples, one panel per platform-topic combination.

The distributions exhibit a predominantly U-shaped pattern, with platform variation. For Gemini,

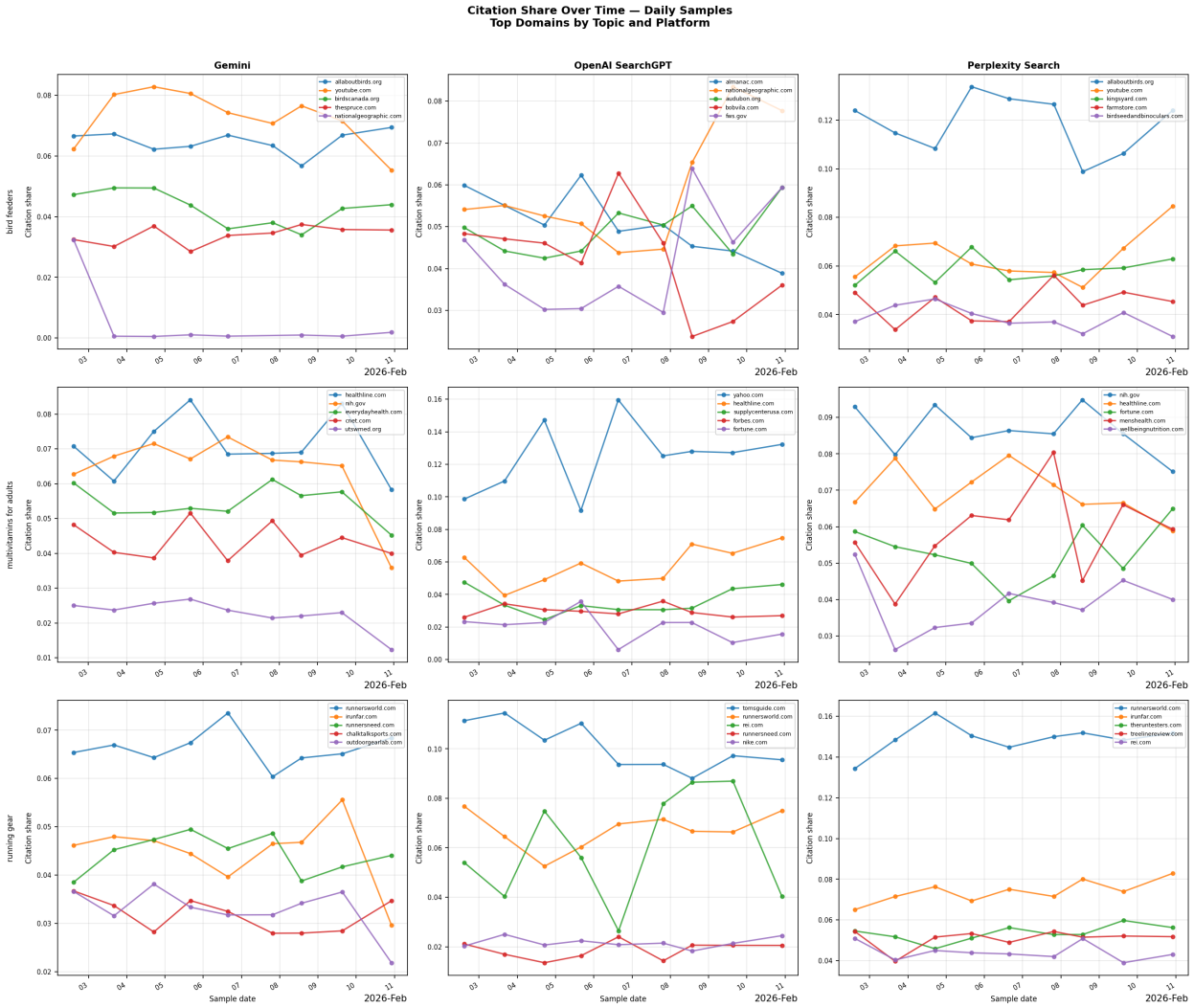


Figure 4: Citation share over time: daily samples. Top domains by topic and platform. 3×3 grid: rows = topics (bird feeders, multivitamins for adults, running gear), columns = platforms (Gemini, SearchGPT, Perplexity). Top 5 domains per panel shown as colored lines. Samples span Feb 3–11, 2026.

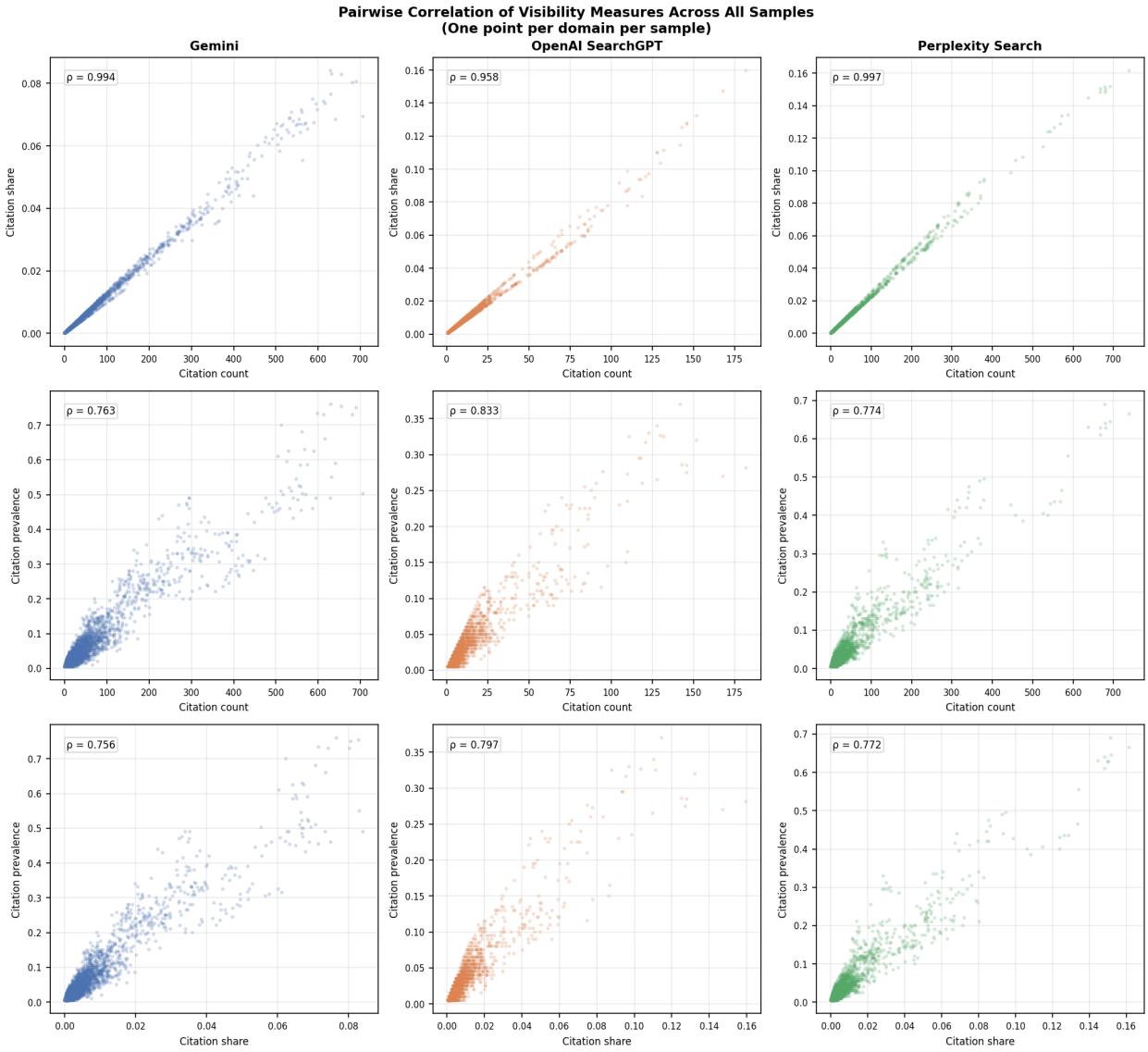


Figure 5: Correlation structure of visibility metrics. 3×3 scatter grid: rows = metric pairs (citation count vs. share; citation count vs. prevalence; citation share vs. prevalence), columns = platforms. One point per domain per sample. Spearman ρ annotated per panel.

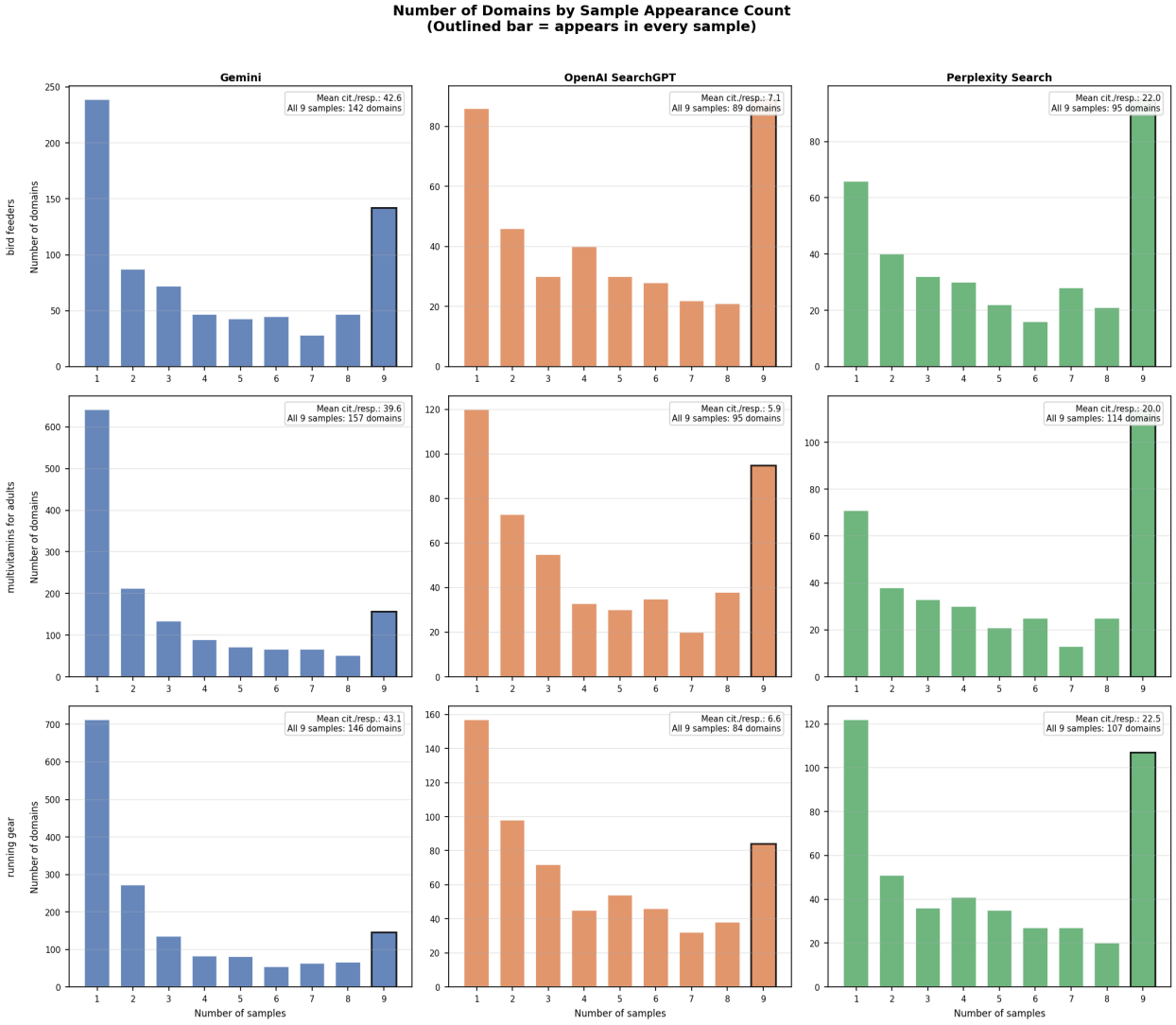


Figure 6: Domain count by number of samples cited. 3×3 grid: rows = topics, columns = platforms. Bar charts showing the number of domains appearing in exactly 1, 2, ..., 9 of the nine daily samples. Outlined bar marks the all-9-samples bin (frequently cited domains).

the U-shape is pronounced: the single-appearance bin dominates (239 domains for bird feeders, 642 for multivitamins, 713 for running gear), middle bins are substantially lower, and the all-nine-samples bin is large and clearly separated (142, 157, and 146 domains, respectively). Proportionally, the gap between the 8-bin and the 9-bin is especially sharp—roughly three times as many domains appear in all nine samples as in exactly eight—providing strong empirical justification for the threshold. For Perplexity, a clear U-shape is also present, with the all-nine-samples bin the largest in two of three topics. For SearchGPT, the distribution is more J-shaped: the single-appearance bin is highest, but the all-nine bin is nearly as high, with a comparatively flat middle.

If our result set had contained only a single sample, the distinction between frequently and infrequently cited domains would have been substantially harder to draw. Both types of domain can appear at any response ordinal during collection. The difference is that frequently cited domains, being structurally embedded in the engine's response distribution, tend to establish themselves early and consistently; infrequently cited domains may appear once and never recur. But in a single sample there is no recurrence data to confirm this, and a domain that appears in the first ten responses of a single run cannot be distinguished from one that would appear in every run. Determining the minimum number of samples, or the minimum number of queries within a single run, required to reliably classify domains as frequently or infrequently cited is an open question directly connected to the minimum sample size guidance identified as a priority in Section 10.

5.4 Power-Law Structure of Citation Distributions

The time series in Section 5.2 illustrate that visibility metrics vary across samples, but illustrating variance is not the same as quantifying it. To quantify uncertainty in citation share and prevalence estimates, we need to understand the underlying distribution from which those observations are drawn. Two properties of that distribution are particularly consequential: its shape, which determines how variance scales with citation share across the domain ranking; and its dispersion, which captures how spread out individual domain shares are across repeated samples. Sections 5.4 and 5.5 characterize each in turn, providing the distributional foundation for the bootstrap confidence intervals in Section 5.6.

Figure 7 plots ranked citation share distributions on a log-log scale for all nine platform-topic combinations. The baseline (first sample) is shown as a thick blue line; all subsequent samples as thin gray lines.

All nine panels confirm a clean power-law structure: citation share decays smoothly across ranks on log-log axes throughout the full domain range. The distribution shape is highly stable across samples: the gray lines cluster tightly around the baseline in all panels, with no sample producing a qualitatively different shape.

The degree of clustering, however, differs by platform in a way that anticipates the convergence findings in Section 5.7. Gemini and Perplexity panels show tight inter-sample agreement throughout the full rank range. SearchGPT panels show noticeably more spread, particularly in the mid-rank region: the gray lines fan out around the baseline rather than overlapping it, indicating that individual domain shares shift more substantially between samples even while the overall distributional shape is preserved. This inter-sample spread is a distributional signature of the same within-sample non-stationarity that produces non-monotonic convergence curves for SearchGPT in Section 5.7—the citation distribution is not merely noisier for SearchGPT but less stable across the query sequence.

Tail lengths differ by platform: Gemini extends to rank approximately 300-400 (consistent with its higher per-response citation volume), Perplexity to rank approximately 100-200, and SearchGPT to rank approximately 100-150. The baseline and subsequent sample curves overlap closely in every

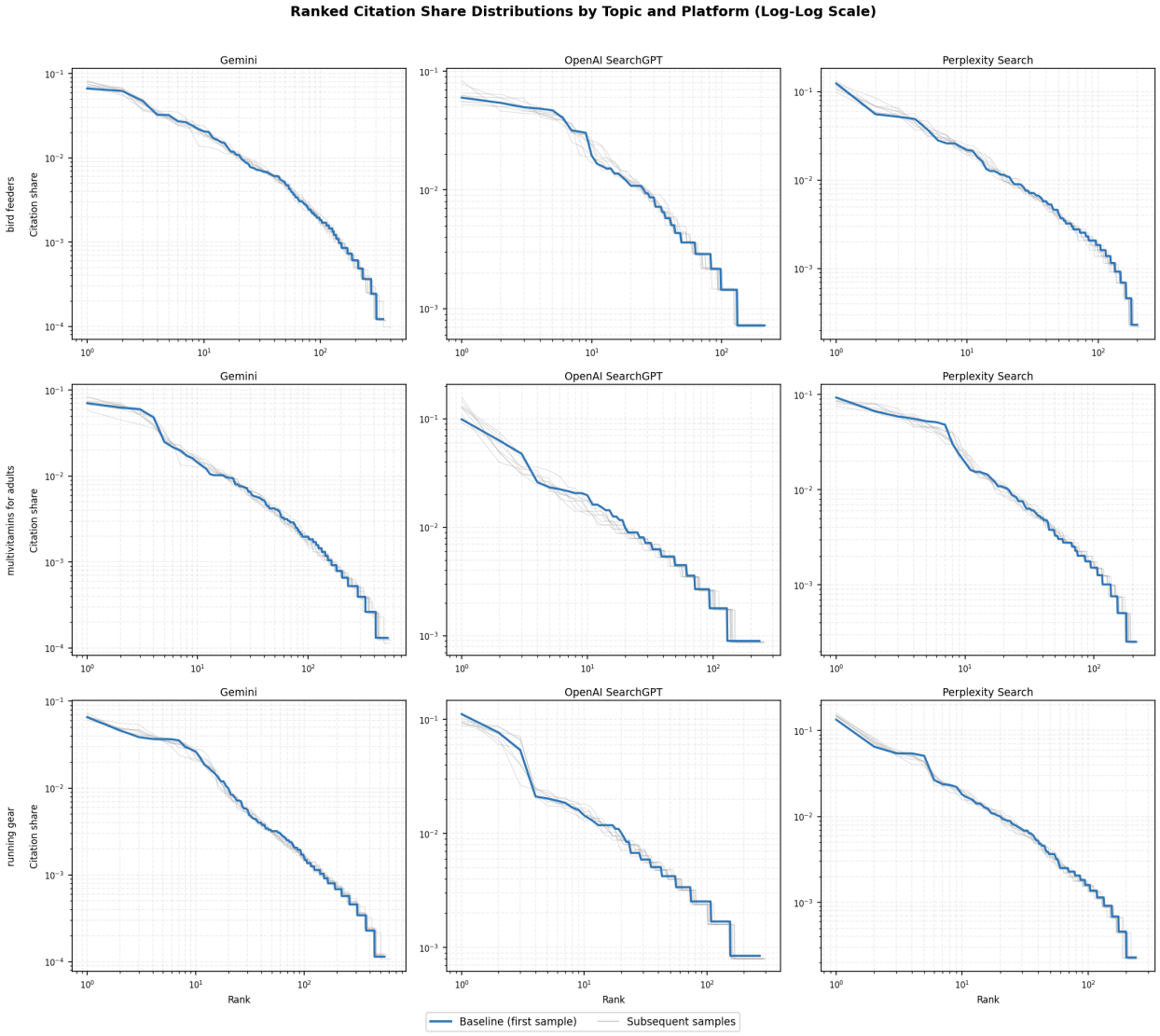


Figure 7: Ranked citation share distributions on log-log scale. 3×3 grid: rows = topics (bird feeders, multivitamins for adults, running gear), columns = platforms. Thick blue line: baseline (first sample). Thin gray lines: subsequent samples. Log-log scale reveals power-law form.

panel, indicating that the distributional shape is stable even when individual domain ranks shift between samples.

The power-law structure has a direct implication for how variability should be measured. Because citation share spans several orders of magnitude across the domain distribution, absolute variance is not scale-invariant: a domain with 10% share has far larger absolute variance than a domain with 0.1% share, even if their relative variability is similar. This motivates the use of log-space dispersion measures in Section 5.5.

Figure 8 overlays the frequently-cited and infrequently-cited domains on the log-log rank distribution, confirming that frequently-cited domains occupy the smooth upper portion of the power-law curve, while tail domains form the scattered lower portion but follow the same distributional shape.

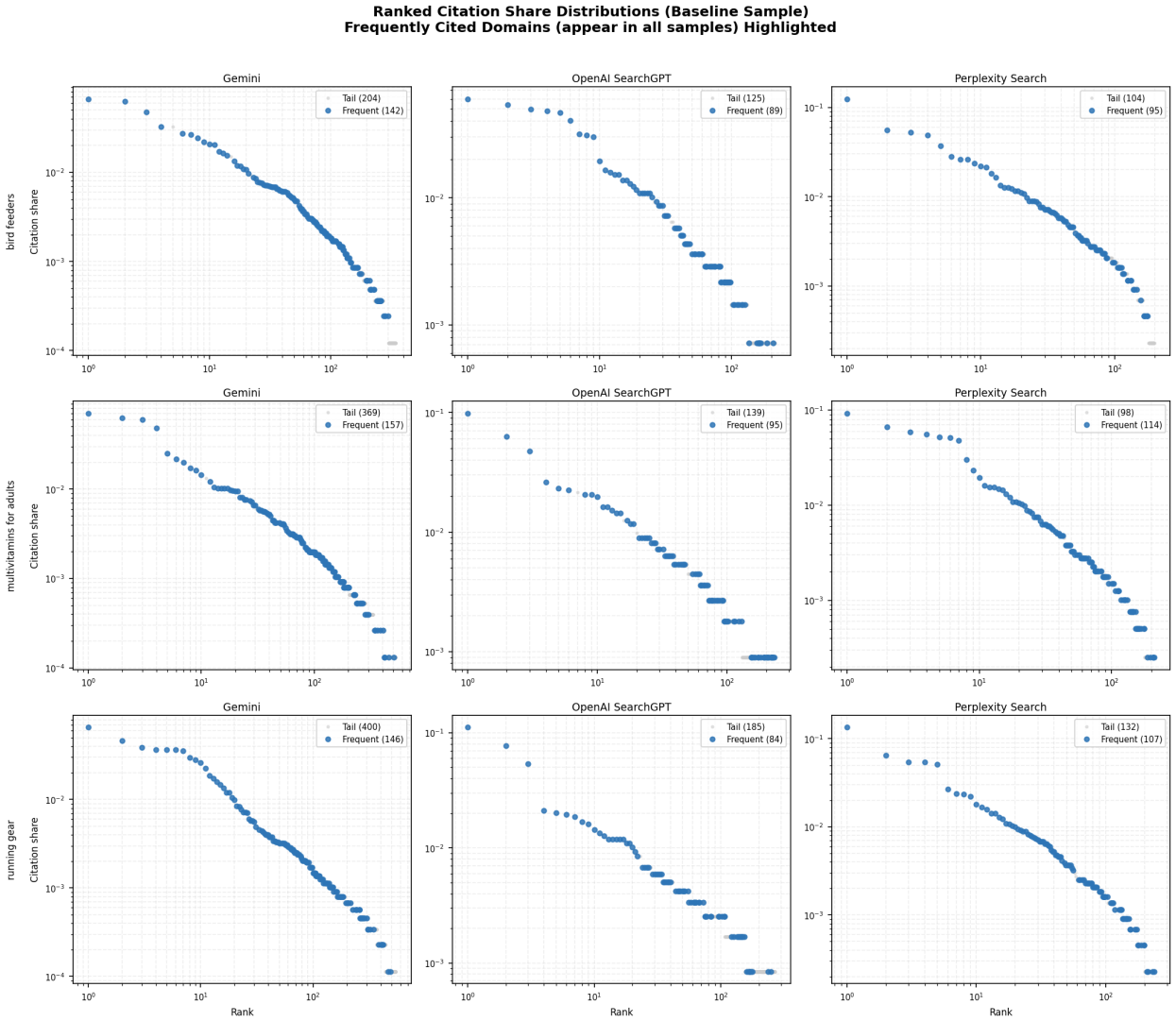


Figure 8: Ranked citation share distributions with frequently-cited domain classification. 3×3 grid: rows = topics, columns = platforms. Baseline sample only. Filled circles: frequently-cited domains (cited in all 9 samples). Faint points: tail domains. Domain counts per panel annotated in legend.

5.5 Log-Space Dispersion

For the set of frequently cited domains, we quantify dispersion using log-space statistics. Because citation share distributions follow a power-law form, variability is multiplicative rather than additive: differences between samples are better characterized as proportional changes than as absolute differences. The primary dispersion measure used throughout is the log-std: the standard deviation of log(citation share) across the nine daily samples, computed per domain. A log-std of 0.5 means that citation share values for a given domain typically deviate by ± 0.5 log units from the geometric mean across samples, equivalent to a multiplicative factor of approximately $\sqrt{e} \approx 1.65$, meaning the share tends to vary between roughly 60% and 165% of its central value. Log-std is scale-invariant: a domain at 1% share and a domain at 10% share with the same log-std are equally variable in relative terms.

Figure 9 summarizes domain-level log-space dispersion across all nine platform-topic combinations.

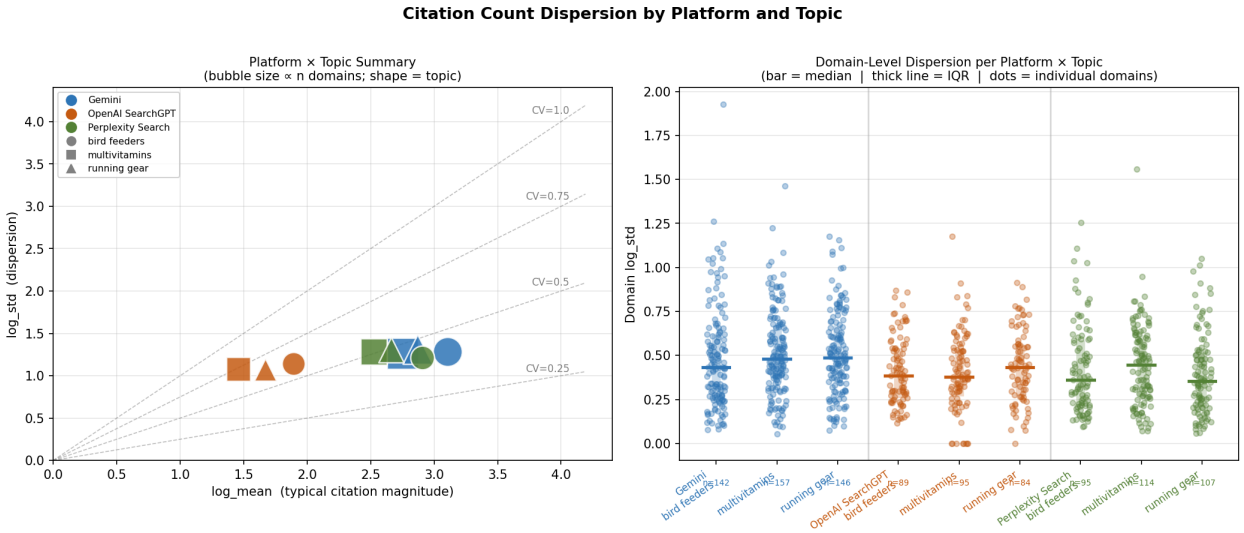


Figure 9: Log-space dispersion summary. Panel A: platform-topic summary of mean and median log-std. Panel B: domain-level log-std strip plot, one strip per platform-topic combination (9 strips).

Table 5: Platform-level log-std summary. Mean and median domain log-std aggregated across all topics and frequently-cited domains, by platform.

Platform	Mean Domain Log-std	Median Domain Log-std
Gemini	0.504	0.469
SearchGPT	0.417	0.397
Perplexity	0.421	0.378

Platforms differ meaningfully in their dispersion profiles, though the ordering is counterintuitive. Gemini has the highest mean domain-level log-std (0.504, median 0.469), followed by Perplexity (mean 0.421, median 0.378) and SearchGPT (mean 0.417, median 0.397). This does not mean Gemini is the hardest platform to measure: it reflects that Gemini has substantially more frequently-cited domains (142-157 per topic) than SearchGPT (84-95) or Perplexity (95-114), and that the larger tail includes many low-share domains cited sporadically, which elevates the platform mean.

For the head of the distribution, the stability ordering reverses. The most stable frequently-cited domains in the dataset are Perplexity's top performers: *runnersworld.com* (log-std = 0.062), *nih.gov* (0.072), *allaboutbirds.org* (0.095). These reflect the structural concentration that makes dominant domains easy to estimate precisely. The least stable domains are tail entries that are cited frequently but erratically: *gardeningknowhow.com* (Gemini bird feeders, log-std = 1.93) and *mindbodygreen.com* (Perplexity multivitamins, log-std = 1.56) are extreme examples—equivalent to citation shares varying by factors of approximately 6× and 3× their geometric mean, respectively.

A notable anomaly appears in SearchGPT multivitamins: nine frequently-cited domains have log-std = 0.0, meaning identical citation counts across all nine jobs. This directly corroborates the platform's Jaccard distribution observed in Section 5.1. SearchGPT has a deterministic layer that fires consistently for certain domain-query pairings, almost certainly domains that are associated with a single query type, which SearchGPT answers identically every time.

The combination of these two properties, a deterministic layer for some domains and high volatility for others, makes SearchGPT qualitatively different from Gemini and Perplexity as a measurement target. For Gemini and Perplexity, higher log-std values reflect quantitative imprecision: more queries reduce uncertainty at a predictable rate. For SearchGPT, the volatility in non-deterministic domains reflects within-sample non-stationarity: the citation distribution shifts across the query sequence in a way that is not simply smoothed away by collecting more queries. This distinction has direct consequences for sample size requirements, which are substantially larger for SearchGPT and in some cases cannot be met within a realistic budget, a finding elaborated in Section 5.7.

5.6 Bootstrap Confidence Intervals

The preceding sections have characterized the structure of citation variability across repeated samples. We now turn to the core inferential question: given a single sample of responses, how can a practitioner quantify the uncertainty in the resulting visibility estimates?

Bootstrap resampling with replacement provides a tractable solution. For a single sample of N responses, we generate B bootstrap replicates by sampling responses with replacement and recomputing the visibility metric for each replicate. The resulting distribution of bootstrap estimates provides an empirical approximation of the sampling distribution of the metric, from which confidence intervals (CI) can be computed. Because citation share is a ratio (domain citations / total citations), both numerator and denominator are recomputed from each bootstrap replicate; resampling at the response level correctly propagates the dependence structure.

Figure 10 presents 95% bootstrap confidence intervals for each topic on each platform.

At a 95% confidence level, the span of the confidence intervals is wide. For most of the frequently cited domains on SearchGPT, the span is from 3 to 6 percentage points. For Gemini and Perplexity, the intervals are narrower but still consequential. The top-ranked domains have the largest citation shares and the largest CI spans: *everydayhealth.com* (multivitamins on Gemini) has a citation share of 6.0% and a CI span of 3.2 percentage points, and *runnersworld.com* (running gear on Perplexity) a citation share of 13.4% and a CI span of 4.7 percentage points. Given the span of these confidence intervals, apparent differences between many of the point estimates—for any topic on any platform—are statistically indistinguishable from sampling noise, as suggested by the many overlapping confidence intervals.

It's worth noting that the bootstrap CI computed from a single sample will not always contain the cross-sample mean. For example, the point estimate for *nationalgeographic.com* (bird feeders on Gemini) is 0.032, which lies far above the cross-sample mean of 0.005, and the CI of 0.024-0.042 fails to capture the long-run value. A practitioner relying on this single sample would rank *nationalgeographic.com* as a top-cited domain on Gemini—a conclusion that all subsequent samples

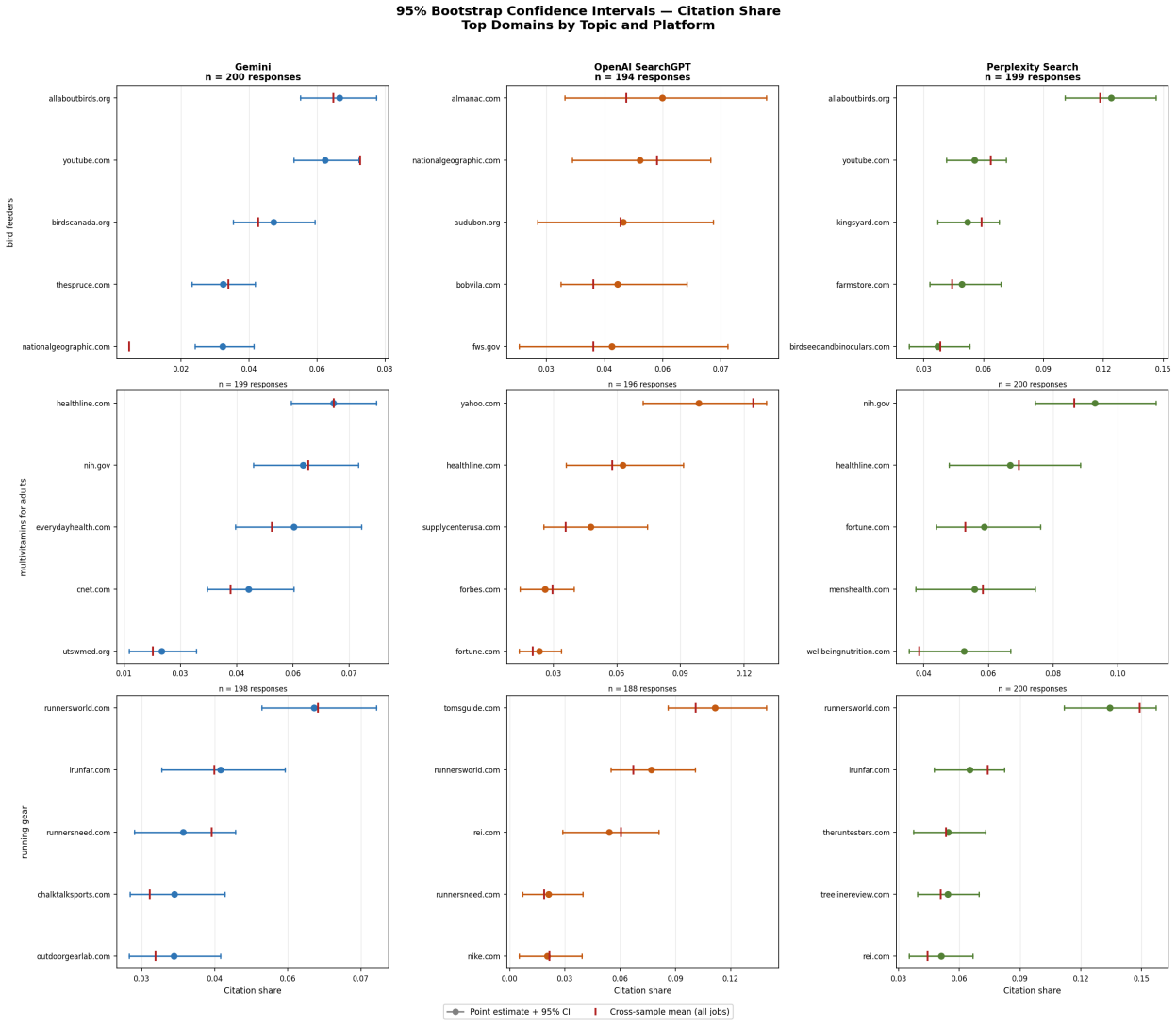


Figure 10: 95% bootstrap confidence intervals for citation share. 3×3 grid: rows = topics, columns = platforms. For each panel, the dots represent the point estimates from the baseline sample, the error bars show the 95% confidence intervals (1,000 bootstrap replicates), and the red tick marks show the cross-sample mean across all nine daily jobs.

contradict. Such misses are not inherently pathological, however: with a 95% confidence interval we expect, by definition, that approximately 1 in 20 estimates will fall outside the interval purely due to sampling variability. However, the magnitude of the deviation in cases like *nationalgeographic.com* illustrates how a single snapshot can still produce highly misleading conclusions about domain visibility.

5.7 Effect of Sample Size on Confidence Interval Width

Figure 11 plots CI width as a function of the number of queries for each platform-topic combination, separately for citation share (left column) and citation prevalence (right column). Each line represents the maximum CI width across all frequently cited domains for a given platform and topic. This reflects a worst-case precision bound: if even one domain has a CI too wide to interpret, the sample cannot reliably distinguish among domain pairs.

The dominant domain typically drives this maximum. Citation shares in the range 0.05-0.15 lie near the middle of the proportion scale, where variance $p(1-p)$ is largest in absolute terms. A domain with a 12% share therefore carries substantially more absolute variance in its citation counts than a domain at 1%, even though it may appear more stable in relative terms.

Each panel overlays a $1/\sqrt{n}$ reference curve representing the CI width expected under ideal conditions: independent, identically distributed responses drawn from a fixed underlying distribution. For a 95% confidence interval on a proportion p, the theoretical width is $3.92 \cdot \sqrt{p(1-p)/n}$, where $3.92 = 2 \times 1.96$ spans both tails of the normal distribution. The reference curves for citation share and citation prevalence are anchored at different vertical scales because prevalence values (0.15-0.45) live closer to 0.5, where $p(1-p)$ is larger, while citation share values (0.03-0.15) lie in a region where $p(1-p)$ is smaller. Both curves have the same $1/\sqrt{n}$ shape; only the scaling constant differs.

We define target CI widths of 0.05 for citation share and 0.15 for citation prevalence as practical precision benchmarks. Deviations from the reference curve are informative in two ways. First, vertical distance from the reference indicates how efficiently additional queries contribute to precision. Curves consistently above the reference suggest that responses are not fully independent. Positive correlation across queries or drift in the citation distribution reduces the effective information contributed by each query. Curves below the reference typically reflect unusually concentrated citation distributions. Second, local bumps, where CI width temporarily increases or stalls before resuming its decline, indicate within-sample non-stationarity: the citation distribution shifts during part of the sample before stabilizing again.

For citation share, Gemini converges fastest across all topics. The Gemini CI width crosses the 0.05 target at approximately $n \approx 30$ for bird feeders and $n \approx 40$-50 for multivitamins and running gear. The decline is smooth and closely tracks the $1/\sqrt{n}$ reference curve, indicating that Gemini's citation distribution is relatively stable within the sample and that queries contribute approximately the expected amount of statistical information.

Perplexity represents the intermediate case. CI widths cross the target at roughly $n \approx 90$-100 across topics. Convergence is less smooth than Gemini's, with visible bumps in the bird feeders and multivitamins curves indicating localized shifts in the citation distribution, but all three topics reach the target within the 200-query sample.

SearchGPT shows the slowest and least monotonic convergence. The bumps visible in all three citation-share curves indicate that the underlying distribution shifts across queries. Although the curves begin to smooth after $n \approx 120$, this should not be interpreted as evidence that collecting 120 or more queries resolves the instability. As n approaches the full sample size of 200, subsamples are drawn from an increasingly exhausted pool, forcing the bootstrap variance to stabilize. Consequently, the curve must flatten as n$\rightarrow$200, regardless of the true volatility of the system. Whether the

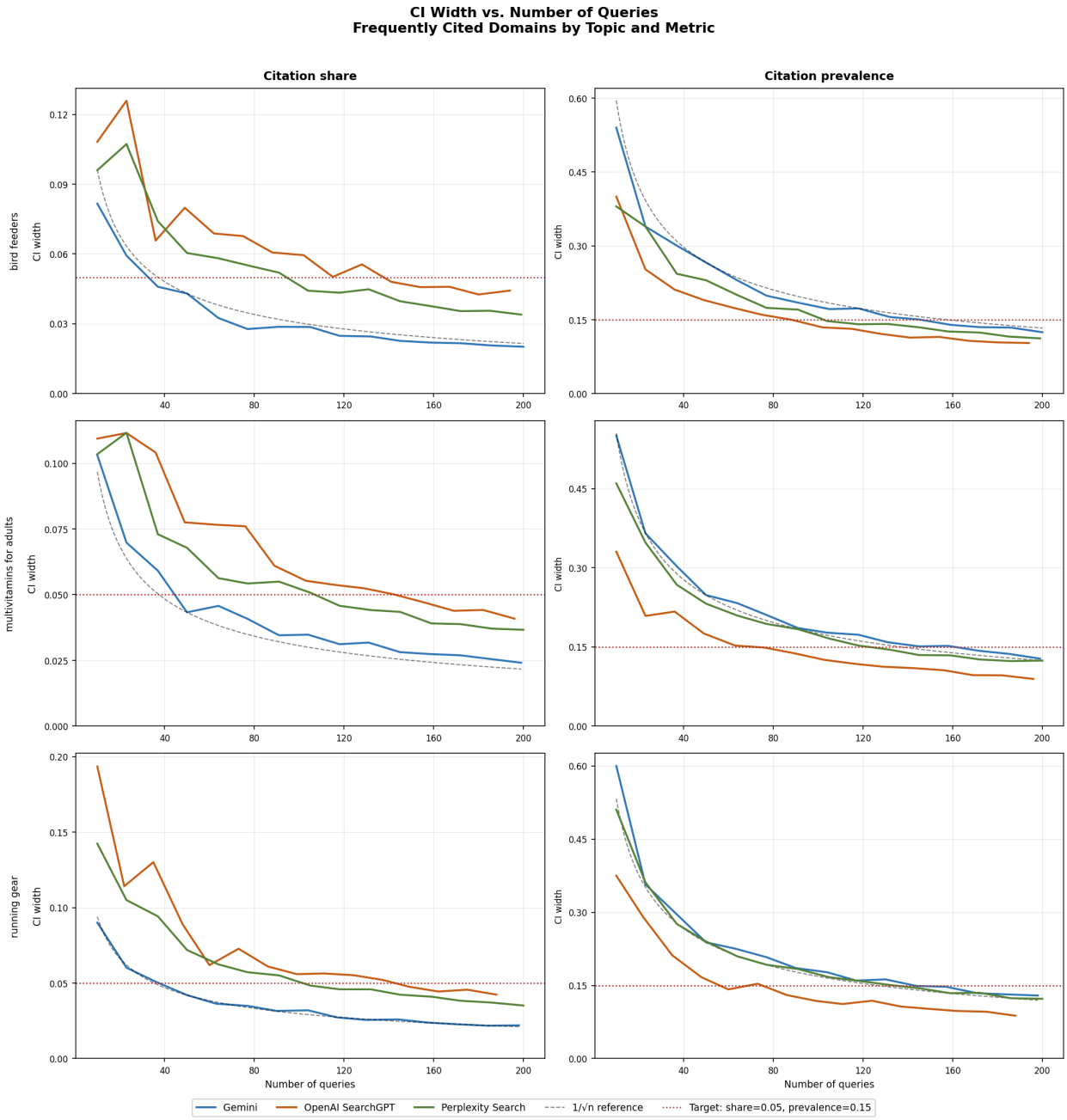


Figure 11: CI width as a function of number of queries. 3×2 grid: rows = topics (bird feeders, multivitamins for adults, running gear), left column = citation share, right column = citation prevalence. One line per platform (Gemini: blue; SearchGPT: orange; Perplexity: green). Gray dashed curve: $1/\sqrt{n}$ reference. Red dotted horizontal line: target width (share: 0.05; prevalence: 0.15).

underlying distribution is stable across repeated collections is therefore a separate question, addressed in Sections 5.4 and 5.5, where SearchGPT's log-std values and daily time series show persistent citation volatility across all nine samples.

For citation prevalence, the platform ordering reverses. SearchGPT converges fastest, crossing the 0.15 target earliest in every topic ($n \approx$ 60-80), followed by Perplexity ($n \approx$ 100-140), and then Gemini ($n \approx$ 140-150). This reversal reflects structural differences between the two metrics: citation share is driven by the absolute volume of citations to dominant domains, whereas prevalence measures how broadly a domain appears across responses, which is easier to estimate for platforms that produce fewer citations per response.

These convergence curves provide practical guidance for minimum sample sizes. For citation share, Gemini requires roughly $n \approx$ 40-50 queries to achieve 95% confidence intervals spanning five percentage points, Perplexity requires about $n \approx 100$, and SearchGPT requires $n \geq 150$, reflecting the increasing degree of non-stationarity in their citation distributions. For citation prevalence, SearchGPT requires $n \approx$ 60-80 queries to achieve intervals spanning fifteen percentage points, whereas Gemini and Perplexity may require nearly twice as many ($n \approx$ 140-150).

Collecting fewer queries than these thresholds results in proportionally wider confidence intervals, making it difficult to distinguish signal from noise. For example, a domain with a true citation share of 8% and a CI spanning ten percentage points cannot be meaningfully distinguished from domains at 3% or 13%; apparent differences or changes remain statistically uninterpretable.

Non-monotonic convergence further complicates interpretation. If CI width declined smoothly with n, practitioners could apply an early-stopping rule: collect queries until the CI width falls below the target and then stop. The SearchGPT curves show why this approach is unsafe. CI width can temporarily narrow and then widen again as additional queries are added, reflecting shifts in the citation distribution. A practitioner who stopped at a fortuitously narrow intermediate point would report a CI that understates the true uncertainty. The safer approach is therefore to commit in advance to a fixed sample size based on prior measurements of the platform and topic, rather than using the running CI width as a stopping criterion.

The convergence curves presented here are a foundation for a more practical question that this paper deliberately leaves open: given a target CI width and a platform, what is the minimum number of queries a practitioner should collect? The platform- and topic-specific crossing points identified in this section—and the non-convergence of SearchGPT for bird feeders citation share—provide the empirical basis for that guidance. Deriving it rigorously, as a function of citation distribution parameters, is reserved for follow-up work and identified as a priority in Section 10.

5.8 Distribution-Wide Rank Stability

The preceding sections quantify measurement uncertainty primarily for the top-5 frequently-cited domains. This is a natural starting point, these are the domains a practitioner is most likely to monitor, but it raises a question about generalizability: is the instability we observe concentrated in the head of the distribution, where the confidence intervals are widest, or does it extend across the full ranking? If rank orderings are unstable throughout the frequently-cited domain set, then the measurement uncertainty problem documented in Sections 5.6 and 5.7 is not a feature of just the top-ranked domains but a property of the entire distribution. That would in turn mean that the value of confidence intervals extends equally to the entire distribution, not just to the top-ranked domains that happen to receive focused attention.

The weighting is a deliberate methodological choice motivated by the noise properties of low-share domains. In an unweighted Spearman, a domain moving from one citation to two citations can jump many rank positions among the hundreds of low-share domains in the frequently-cited

set. Because such rank changes are numerous and driven by single-citation fluctuations rather than structural shifts, an unweighted correlation has high bootstrap variance: the CI for each pair becomes very wide, and many pairs are classified as insufficient (CI width > 0.25) even when the rank ordering of the commercially relevant head of the distribution is highly stable. Weighting by citation share concentrates the correlation on rank changes that are substantively meaningful, damping the noise from the low-share tail without discarding it entirely. The result is a measure that speaks to distribution-wide stability in a way that is both statistically tractable and interpretively appropriate: a rank swap between the 1st and 5th domain is correctly treated as more consequential than a rank swap between the 200th and 201st.

Confidence intervals for each pair's ρ are constructed by bootstrapping over domains: the set of frequently-cited domains is resampled with replacement, and the weighted Spearman correlation is recomputed on each resample. The 95% CI is the interval between the 2.5th and 97.5th percentiles of the resulting bootstrap distribution across 1000 iterations. This is a standard percentile bootstrap, without bias correction or Fisher z-transformation.

A consequence of this construction is that the point estimate can occasionally fall outside the CI band. Two properties interact to produce this. First, the bootstrap distribution of weighted ρ is skewed when the point estimate is close to 1: because ρ is bounded above by 1, bootstrap replicates are constrained to produce correlations no greater than 1, compressing the upper tail and pulling the lower percentile upward. Second, resampling domains with replacement redistributes the weights (e.g., a domain drawn twice receives double weight) changing the effective contribution of each domain to the correlation relative to the point estimate computed on the full, naturally-weighted set. The combination of boundary compression and weight redistribution can produce a bootstrap distribution whose lower percentile exceeds the observed point estimate. This is a known limitation of the percentile bootstrap near parameter boundaries and does not indicate a computational error. It arises precisely in the cases where stability is highest, and is therefore conservative in the direction of the paper's argument.

Figure 12 plots rank correlation across consecutive job pairs for all nine platform-topic combinations, with the 0.9 threshold shown as a red dashed line, gray shading for insufficient pairs (where the CI width exceeds 0.25), and a yellow diamond for the span comparison (first job vs. last job). Table 6 summarizes the results numerically.

Table 6: Distribution-wide rank stability summary, one row per platform-topic combination. Pairs = total consecutive job pairs; Sufficient = pairs with bootstrap CI width ≤ 0.25; Stable = sufficient pairs with Spearman $\rho \geq 0.9$; Mean Rank ρ and Mean CI Width are averaged over sufficient pairs; Span Rank ρ compares the first and last job; Span Drift indicates whether a statistically and practically significant drift was detected over the nine-day window.

Platform	Topic	Pairs	Sufficient	Stable	Mean Rank ρ	Mean CI Width	Span Rank ρ	Span Drift
Gemini	bird feeders	8	8	4	0.866	0.150	0.689	TRUE
Gemini	multivitamins	8	8	2	0.806	0.209	0.726	TRUE
Gemini	running gear	8	8	7	0.913	0.127	0.900	TRUE
SearchGPT	bird feeders	8	5	0	0.905	0.225	0.744	FALSE
SearchGPT	multivitamins	8	0	0	—	—	0.677	FALSE
SearchGPT	running gear	8	0	0	—	—	0.727	FALSE
Perplexity	bird feeders	8	8	0	0.882	0.163	0.850	TRUE
Perplexity	multivitamins	8	1	1	0.959	0.250	0.947	FALSE
Perplexity	running gear	8	8	0	0.872	0.163	0.864	TRUE

The results differ sharply by platform and topic. Gemini running gear is the cleanest stability

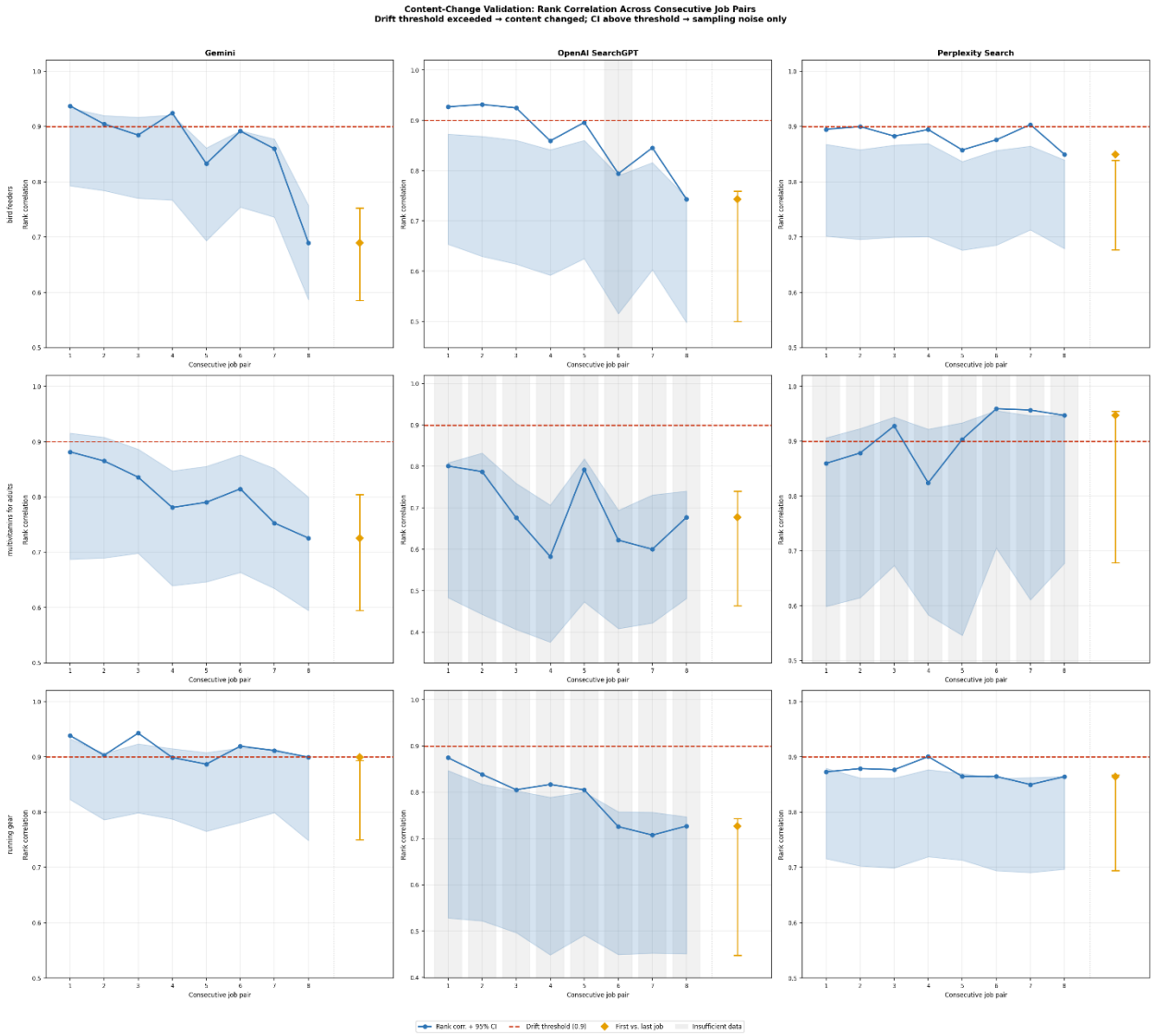


Figure 12: Distribution-wide rank stability: rank correlation across consecutive job pairs. 3×3 grid: rows = topics (bird feeders, multivitamins for adults, running gear), columns = platforms (Gemini, SearchGPT, Perplexity). Blue line with 95% CI band: Spearman rank correlation for each consecutive job pair. Red dashed line: stability threshold ($\rho = 0.9$). Gray shading: insufficient data pairs (CI width > 0.25). Yellow diamond: span comparison (first vs. last job).

result in the dataset: all eight pairs are sufficient, seven are stable, mean $\rho = 0.913$, and the span comparison $\rho = 0.900$—right at the threshold. Gemini bird feeders and multivitamins tell a different story: all eight pairs are sufficient in both cases, but only four and two pairs, respectively, are stable, mean ρ falls to 0.866 and 0.806, and span ρ drops to 0.689 and 0.726—well below the threshold. Span drift is detected for all three Gemini topics, indicating that cumulative drift over the nine-day window is a platform-level property for Gemini regardless of topic.

SearchGPT presents the most extreme picture. Bird feeders has five sufficient pairs but zero stable ones, with a mean CI width of 0.225—close to the sufficiency boundary. Multivitamins and running gear have zero sufficient pairs entirely, making mean ρ and mean CI width undefined; span ρ values of 0.677 and 0.727 can be computed but carry no inferential weight given the insufficient pair counts. No span drift is detected for SearchGPT, but this is an artefact of the insufficient data rather than evidence of stability—there is simply not enough precision to detect drift that may be present.

Perplexity occupies a middle position. Bird feeders and running gear each have all eight pairs sufficient and span ρ values of 0.850 and 0.864, with span drift detected in both. Multivitamins has only one sufficient pair—the mean CI width of 0.250 confirms it sits at the boundary of the sufficiency criterion—but that one pair is stable and span $\rho = 0.947$. The wide CI bands for Perplexity reflect head-domain concentration rather than instability per se: a distribution dominated by one or two high-share domains has fewer effective rank positions to correlate, inflating CI width.

An important cross-cutting finding: short-window consecutive-pair stability does not imply long-window stability. For Gemini bird feeders and multivitamins, each consecutive pair has $\rho > 0.80$ with CIs straddling the threshold, yet the span comparison falls to approximately 0.69-0.73—a cumulative drift that would be invisible from any single adjacent-pair comparison. Long-window span analysis is therefore a necessary complement to consecutive-pair monitoring.

The rank stability results confirm that distribution-wide instability is the norm rather than the exception across platforms and topics. For the combinations where rank correlation falls below the sufficiency threshold, Gemini bird feeders and multivitamins across the nine-day window, and SearchGPT across most combinations, uncertainty is not confined to individual domain shares at the head of the distribution but propagates through the entire ranking. Confidence intervals are therefore not a refinement useful only for comparing the top two or three domains: they are a necessary tool for any inference drawn from a citation distribution, at any rank position.

The sufficiency criterion ($\rho \geq 0.9$) used here is a practical benchmark rather than a formally derived threshold. Whether this criterion is appropriate, and how it should be calibrated as a function of the number of frequently-cited domains and the intended use of the ranking, is an open question. Establishing principled sufficiency thresholds and connecting them to minimum sample size guidance is a priority for future work identified in Section 10.

6 Content-Change Validation

The findings in Section 5 document substantial citation variability across platforms, topics, and sampling occasions. A natural alternative explanation is that the content of cited pages changed between samples, making some pages more or less relevant and thereby altering citation probabilities. This section reports a methodological control designed to rule out that explanation.

6.1 Approach

Using the content-change monitoring infrastructure described in Section 4.5, we identify domains for which citation counts changed significantly between adjacent daily samples. For each flagged domain,

we compare the SHA-256 checksums of human-readable page content across the corresponding job pairs. A stable checksum indicates that the page content did not change; a changed checksum indicates a potential content update.

The drift detection procedure uses the first daily sample as a frozen baseline and compares each subsequent sample against it using a chi-squared test with both statistical and practical significance thresholds. A domain is flagged if both (a) the chi-squared p-value falls below 0.05 and (b) the total share delta exceeds the practical significance threshold.

Content status for each domain-job transition is classified as follows. A transition is classified as unchanged if the content hash is identical to the previous job. It is classified as changed if at least one URL for the domain has a different hash from the previous job. It is classified as unknown if hash data are unavailable for this or the preceding job. By construction, the first job in any panel receives unknown status, since there is no prior job to compare against.

6.2 Results

Figure 13 plots citation share over time for the top-5 frequently-cited domains per panel, with each data point shaped to indicate the content status of that domain at that job relative to the previous job.



Figure 13: Citation share over time—content-change status. 3×3 grid: rows = topics, columns = platforms. Top-5 frequently-cited domains per panel. Daily samples. Marker shapes encode content status: circle = unchanged (hash identical to previous job); square = changed ($\geq$1 URL hash differs); triangle = unknown (no hash data for this or previous job). Triangles are confined to the first job in every panel.

Triangle markers (unknown) are confined to the first job in every panel, confirming that hash coverage is complete from the second job onward. The overwhelming majority of data points are circles, indicating that the underlying pages were stable across consecutive job transitions for most domains throughout the nine-day window. The citation share volatility documented in Sections 5.4, 5.5, and 5.7 therefore cannot be attributed to changes in cited page content: the oscillations, rank crossings, and platform-level differences occur against a background of largely stable source material.

A subset of domains display square markers on every job transition after the baseline. The most prominent example is *runnersworld.com* on Perplexity (running gear topic), which shows a single unknown triangle on the first job followed by changed squares on all subsequent jobs. This pattern is consistent with server-side rendering variation: access timestamps, session tokens, advertisement slots, or similar ephemeral page elements that produce a different byte sequence on each HTTP request without altering human-readable content. Hash-based change detection is a conservative signal: identical hashes definitively rule out content change, but differing hashes are a necessary, not sufficient, condition for meaningful editorial change. We report these cases transparently and do not attempt to distinguish rendering variation from genuine content change.

The substantive finding is unaffected by these ambiguous cases. Even for domains where hash-based stability cannot be confirmed, citation share volatility of the magnitude observed here (rank crossings within a single four-hour high-frequency window, CI widths of 3-7 percentage points on daily data) cannot plausibly be explained by incremental server-side changes that produce hash churn. The variability is structural, not content-driven.

7 Discussion

7.1 Implications for Visibility Measurement

The central implication is that single-run measurements of domain visibility in generative search cannot be interpreted as reliable estimates of an underlying true citation rate. The confidence intervals reported in Section 5.6 show that observed values for a given domain can vary substantially across repeated samples, and that many apparent differences between domains are not statistically distinguishable. This has concrete consequences for how practitioners should interpret visibility measurements. Ranking changes between samples, apparent surges or declines in citation share, and comparisons between domains that are close in the distribution may all be artifacts of sampling noise rather than signals of genuine change.

The appropriate inferential unit is not the point estimate but the confidence interval. A domain with citation share 6% $\pm$ 2 percentage points (95% CI: 4%-8%) is in a meaningfully different measurement state than one reported as simply "6%." The former acknowledges the uncertainty inherent in the measurement; the latter conveys false precision. Bootstrap confidence intervals, computed at the response level using the procedure described in Section 5.6, provide this information with minimal additional computational cost.

The rank stability analysis in Section 5.8 extends this implication beyond the top-ranked domains. A practitioner might reasonably suppose that measurement uncertainty is primarily a problem for closely-ranked domains near the head of the distribution, and that lower-ranked domains can be safely ignored or that their relative ordering is stable. The distribution-wide rank correlation results contradict this: for most platform-topic combinations, the rank ordering of the full frequently-cited domain set is either unstable across samples or cannot be assessed with sufficient precision. Confidence intervals are therefore not a refinement applicable only to the top two or three

domains—they are a necessary tool for any inference drawn from a citation distribution at any rank position.

7.2 Implications for Visibility Optimization Research

Work in the GEO tradition (Aggarwal et al.) reports visibility improvements from content interventions without confidence intervals. The findings of this paper suggest that such claims require statistical validation: an observed improvement in citation share must be shown to exceed the noise floor of the measurement process before it can be attributed to the intervention. Bootstrap confidence intervals on both the baseline and post-intervention measurements provide the necessary inferential framework.

This requirement is not merely theoretical. Given the CI widths documented in this paper, an intervention that improves a domain's SearchGPT citation share from 8% to 11% cannot be attributed to the intervention with confidence: the improvement of 3 percentage points falls within the typical CI width for a SearchGPT domain. Only larger effects, or effects validated by repeated pre/post measurement, are distinguishable from noise.

7.3 Platform Differences

The three platforms studied here differ in several empirically documented ways that affect both measurement design and interpretation. Citation volume differs by roughly six times across platforms: Gemini produces approximately 40-43 citations per response, Perplexity approximately 20-22, and SearchGPT approximately 6-7. This drives Gemini's larger frequently-cited domain set (142-157 per topic) and longer power-law tail relative to SearchGPT (84-95) and Perplexity (95-114).

Rank stability differs: Perplexity's top-cited domains have the lowest domain-level log-std in the dataset (*runnersworld.com*: 0.062; *nih.gov*: 0.072; *allaboutbirds.org*: 0.095), while Gemini's tail frequently-cited domains include extreme outliers (*gardeningknowhow.com*: log-std = 1.93). Response consistency differs: SearchGPT shows a bimodal Jaccard distribution (either near-perfect repeat or complete divergence) and nine domains with log-std = 0.0 (deterministic citation behavior for those domains), while Gemini and Perplexity show unimodal distributions centered around Jaccard 0.30 and 0.50, respectively.

These differences have direct implications for measurement design. The number of queries needed to achieve a given CI width is platform- and topic-specific. The interpretation of single-sample estimates carries different risks depending on whether the platform's citation behavior is primarily deterministic, stochastic, or intermediate. A measurement protocol designed for Gemini (where n = 40-50 might be sufficient) is inadequate for SearchGPT, where no fixed-n protocol achieves the target precision within a realistic budget. For SearchGPT, the problem is not only that more queries are needed but that within-sample non-stationarity—evidenced by the non-monotonic convergence curves in Section 5.7—means the citation distribution itself shifts across the query sequence. This makes early-stopping rules unreliable and requires committing to a fixed n based on prior knowledge of the platform. Rank stability compounds the picture: SearchGPT multivitamins and running gear have zero sufficient pairs in the rank correlation analysis, meaning the distribution-wide rank ordering cannot be assessed at all with the current sampling budget.

8 Limitations

We identify the following limitations of this study.

- Three topics. The study covers three consumer product topics chosen to represent varied market structures. Generalization to other domains, including B2B topics, navigational queries, and rapidly evolving news topics, requires further study.
- LLM-generated queries. Queries were generated by ChatGPT, which may have implicit associations between topics and sources that shape the query distribution. This is a departure from using observed user query logs and may affect the representativeness of the query set.
- Gemini exclusion from high-frequency regime. Google Gemini was excluded from the ten-minute interval experiment due to API rate limits. The high-frequency findings therefore cover only Perplexity Search and OpenAI SearchGPT.
- Scraping coverage. Content checksums were computed from HTML scraped during data collection. Coverage is incomplete due to server-side blocking (Cloudflare, CAPTCHA) and format limitations (video content, PDFs). Conclusions about content stability are based on the subset of URLs for which scraping succeeded.
- Single observation window. All data were collected over a period of approximately nine days. Longer-term dynamics, including seasonal variation and major platform updates, are not captured.
- Infrequently cited domains. The statistical treatment of zero-inflated citation data is out of scope for this paper and represents an open methodological challenge.

9 Conclusion

Generative search engines are non-deterministic systems, and the citation visibility metrics derived from their outputs are consequently random variables subject to both system-level stochasticity and measurement uncertainty. This paper has demonstrated empirically that this variability is large enough to materially affect analytical conclusions: many apparent differences between domains, and many apparent changes across samples, are not statistically distinguishable from noise.

Bootstrap confidence intervals provide a principled and tractable method for quantifying this uncertainty. Applied to citation count, share, and prevalence metrics, they reveal that single-run measurements carry substantial imprecision that is invisible when results are reported as point estimates. The intervals are wide enough to render common analytical claims (e.g., domain A outranks domain B; visibility improved after intervention X) untestable without repeated sampling.

The content-change validation confirms that observed variability is predominantly structural: the underlying source pages are largely stable, and the citation oscillations reflect engine behavior rather than changing content. The power-law characterization of citation distributions provides a theoretical grounding for log-space dispersion analysis and motivates the distinction between frequently and infrequently cited domains that is central to the measurement framework. The distribution-wide rank stability analysis confirms that this measurement uncertainty is not confined to the top-ranked domains: rank instability extends across the full frequently-cited domain set for most platform-topic combinations, establishing that confidence intervals are necessary for inference at any rank position, not merely among the most visible domains.

We hope these findings motivate a shift in how visibility measurement is practiced and reported in the growing literature on generative search optimization toward a framework that treats citation visibility as what it is: a probabilistic quantity estimated from a finite, noisy sample.

10 Future Work

The following directions are identified as natural continuations of this work.

- Minimum sample size guidance. A follow-up study will provide principled guidance on how many queries are needed to achieve a target confidence interval width for citation share and prevalence metrics, as a function of citation count distribution parameters and the platform being measured.
- Statistical treatment of infrequently cited domains. Zero-inflated count models and occupancy models from ecology offer potential frameworks for estimating citation probabilities for domains that appear in only some samples. Whether platform-level experimentation with grounding sources contributes to the observed presence/absence pattern is an empirical question for future study.
- Longer observation windows. A study spanning weeks or months would enable characterization of longer-term dynamics, including seasonal patterns and the effects of platform architecture updates.
- Additional topics and query types. Extending the analysis to B2B, navigational, and news-oriented topics would test the generalizability of the findings.
- Real user queries. Replacing LLM-generated queries with observed user query logs would eliminate the query generation confound and improve ecological validity.

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A Citation Format Examples

The following tables illustrate how citations are implemented in Perplexity and SearchGPT. Both examples are responses to the query, "top rated multivitamins for adults by experts."

B Per-Sample Summary Statistics

Table A.1: Perplexity citation example. Response to the query "top rated multivitamins for adults by experts.

Response
Based on expert reviews for 2026, the **top-rated multivitamins for adults** vary by category and specific needs: **Overall Best Multivitamins. Performance Lab® NutriGenesis® Multi** is recommended as the best overall multivitamin supplement, offering 27 vitamins and minerals in gender-specific formulations for men and women with clean, plant-based delivery [1]. **Ritual Essential for Men** is rated as the best overall multivitamin specifically for men [3], while **Inessa Advanced multivitamin** (£29.99) earns top marks as a premium option with a 5/5 star rating [2]. **Top-Rated by Category.** *Best for meeting daily recommended values:* Swallow daily multivitamin (5/5 stars) [2]. *Best one-a-day:* HealthSpan Gold A-Z multivitamin (5/5 stars) at 13p per day [2]. *Best for women:* Dr Vegan Women's ProMulti (5/5 stars) [2]. *Best all-rounder capsule:* Viridian High-5 (4.5/5 stars) [2]. *Best budget option:* Healthspan Multivitality Pro (4/5 stars) at 19p per day [2]. For those over 50, **Centrum Silver Adults 50+** and **Thorne Multi 50+ Daily** are recommended by experts [3][4]. [1] `https://www.performancelab.com/blogs/multi/best-multivitamin` [2] `https://www.bbcgoodfood.com/review/best-multivitamins` [3] `https://www.menshealth.com/health/a19545998/best-multivitamins-for-men/` [4] `https://www.cspi.org/article/multiple-choice-how-find-best-multivitamin`

Table A.2: SearchGPT citation example. Response to the query "top rated multivitamins for adults by experts.

Response
Selecting the right multivitamin can be challenging due to the multitude of options available. Based on expert reviews and evaluations from 2026, here are some top-rated multivitamins for adults: **Thorne Basic Nutrients 2/Day.** A comprehensive multivitamin providing 21 essential vitamins and minerals, including all eight B vitamins and vitamins A, C, D, E, and K. Notably, it offers 250% of the daily value for vitamin D. (health.yahoo.com — `https://health.yahoo.com/wellness/nutrition/vitamins-supplements/article/best-multivitamin-for-2025-202710445.html`) **Nature Made Multivitamin For Him.** An affordable, one-a-day multivitamin tailored for men, containing 22 essential nutrients, including vitamin D, zinc, and selenium. USP-verified for purity and potency. (health.yahoo.com — `https://health.yahoo.com/wellness/nutrition/vitamins-supplements/article/best-multivitamin-men-211005605.html`) **Perelel Women's Daily Trio.** A multivitamin designed specifically for women, offering a blend of essential nutrients to support overall health. (health.yahoo.com — `https://health.yahoo.com/wellness/nutrition/vitamins-supplements/article/best-multivitamin-for-2025-202710445.html`) **Nordic Naturals Men's Multivitamin.** Tailored for men's health, this multivitamin includes saw palmetto and nettle extract to support prostate health, along with essential nutrients like vitamin D and B vitamins. Third-party tested. (health.yahoo.com) **Garden of Life Organics Women's Once Daily.** A USDA-certified organic multivitamin formulated for women, providing over 17 essential nutrients sourced from whole foods. (health.yahoo.com) When choosing a multivitamin, consider factors such as your specific health needs, dietary restrictions, and lifestyle. It is advisable to consult with a healthcare professional to determine the most suitable option for you.

Table B.1: Gemini per-sample summary statistics.

				Citations				
Platform	Topic	Sample	Responses	Mean	Median	Std	Min	Max
Gemini	bird feeders	1	200	41.1	36.0	21.3	9	110
		2	200	42.4	36.0	22.3	7	130
		3	198	39.9	35.0	20.7	7	154
		4	200	42.8	36.0	23.2	7	147
		5	200	41.3	36.0	21.8	10	112
		6	194	40.9	35.0	22.2	11	158
		7	200	41.1	36.5	21.0	10	112
		8	198	42.2	36.5	22.0	6	104
		9	199	51.1	44.0	27.5	7	152
	multivitamins for adults	1	199	38.1	35.0	17.4	3	109
		2	198	40.2	38.0	18.3	3	100
		3	197	39.7	35.0	20.0	9	138
		4	199	37.6	35.0	20.1	7	169
		5	199	40.7	36.0	20.9	5	142
		6	199	38.9	36.0	18.9	3	106
		7	199	38.8	35.0	17.9	8	92
		8	199	38.3	35.0	18.1	5	112
		9	198	44.1	41.5	23.9	8	197
	running gear	1	198	44.4	41.0	22.0	6	136
		2	200	42.4	39.0	21.3	6	132
		3	200	43.4	39.5	21.3	13	134
		4	199	44.0	41.0	21.4	13	148
		5	199	42.1	38.0	19.7	8	117
		6	199	42.0	40.0	19.4	1	94
		7	197	41.9	39.0	18.5	5	108
		8	199	40.6	39.0	18.0	3	100
		9	200	46.8	42.0	26.0	10	202

Table B.2: SearchGPT per-sample summary statistics.

				Citations				
Platform	Topic	Sample	Responses	Mean	Median	Std	Min	Max
SearchGPT	bird feeders	1	194	7.1	7.0	2.6	1	15
		2	195	7.1	6.0	2.8	1	15
		3	196	7.1	7.0	2.6	1	16
		4	198	7.0	6.0	2.6	1	16
		5	197	7.0	6.0	2.5	1	14
		6	197	7.1	7.0	2.5	1	14
		7	197	6.8	6.0	2.8	1	16
		8	198	7.2	6.0	2.7	1	17
		9	194	7.3	7.0	2.9	1	18
	multivitamins for adults	1	196	5.7	5.0	1.9	1	11
		2	197	5.9	5.0	2.2	1	13
		3	193	5.9	5.0	2.1	1	15
		4	196	5.8	5.0	1.9	1	11
		5	192	5.9	5.0	2.2	1	15
		6	189	6.0	5.0	2.1	1	13
		7	195	5.9	5.0	1.8	1	12
		8	194	5.9	5.0	2.0	1	12
		9	193	6.0	5.0	2.0	1	12
	running gear	1	188	6.3	5.0	2.6	1	18
		2	186	6.7	5.0	2.6	1	14
		3	185	6.8	6.0	2.5	1	18
		4	186	6.2	5.0	2.3	1	14
		5	187	6.7	5.0	2.5	1	16
		6	191	6.6	5.0	2.5	1	14
		7	187	6.7	6.0	2.6	1	17
		8	192	6.6	5.0	2.7	1	19
		9	191	6.6	5.0	2.6	1	16

Table B.3: Perplexity per-sample summary statistics.

				Citations				
Platform	Topic	Sample	Responses	Mean	Median	Std	Min	Max
Perplexity	bird feeders	1	199	21.8	21.0	9.4	3	57
		2	200	22.9	22.0	10.2	7	72
		3	200	22.0	20.0	9.7	3	48
		4	200	21.4	20.0	9.7	5	55
		5	200	22.1	20.0	9.9	4	56
		6	200	21.9	20.0	9.9	3	49
		7	199	22.7	21.0	9.8	4	63
		8	200	21.5	21.0	9.4	2	48
		9	200	21.9	20.0	9.5	7	58
	multivitamins for adults	1	200	19.9	18.0	8.5	5	51
		2	200	20.0	19.0	8.5	6	45
		3	200	20.3	19.0	7.8	6	44
		4	200	20.1	18.0	8.3	6	49
		5	200	19.8	19.0	8.1	5	46
		6	200	19.6	19.0	8.0	5	41
		7	200	20.0	19.0	8.2	7	42
		8	199	20.1	19.0	8.4	6	52
		9	200	20.5	18.5	9.0	3	49
	running gear	1	200	21.9	21.0	8.4	5	53
		2	200	22.5	22.0	8.2	5	44
		3	200	22.9	22.0	8.6	7	56
		4	199	22.3	22.0	9.3	4	50
		5	200	22.0	22.0	8.7	4	48
		6	199	22.8	22.0	9.6	4	47
		7	200	22.8	22.0	8.7	7	48
		8	200	22.9	23.0	8.4	5	48
		9	200	22.4	22.0	8.8	6	49